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VISUALISING SPATIAL SOUND IN AUGMENTED REALITY: EVALUATING VISUALISATION TECHNIQUES FOR SOUND SOURCE LOCALISATION

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Abstract

Hearing is an important aspect of our surroundings however it is often impaired meaning information about one's environment is lost. To help regain awareness of a user's aural environment, a prototype system was built which uses augmented reality to visualise the location of sounds sensed by a microphone array. Informed by existing visualisations in Augmented Reality navigation and out of view object visualisation work, three visualisations were created along with their pair combinations. These visualisations were evaluated in order to understand what constitutes an effective visualisation for conveying spatial sound and if combinations of visualisation techniques were applicable. This evaluation involved participants localising sounds in a simulated environment, performance metrics along with surveys and interviews were used to better understand each participants use of visualisations and their preference. It was found that preference amongst visualisations was varied but overall head-mounted explicit visualisations such as the Radar visualisation offer the most consistent feedback across varying distances from sound sources. It was also found that combinations were effective, with one combination being preferred by the highest proportion of participants and aiding in accurate sound source localisation. From our findings, design guidelines were proposed to inform the design of effective future visualisations of sound, enabling those to regain awareness of their aural environment.

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1 | Introduction

Sound is an important aspect of one's environment, providing critical information about their surroundings. Humans rely on hearing to detect, localise, and identify sounds, informing them of their surroundings such as potential dangers, locations of objects, and presence of people (Desai and Mehendale 2022).

When one's hearing is impaired their ability to detect, localise, and identify sounds is reduced (Fostick and Fink 2021). This results in them losing information about their surroundings diminishing their situational awareness. Many situations arise where one's hearing is impaired, surrounding aural information is lost, and situational awareness is reduced (Daniel 2007; Fostick and Fink 2021). For example, there exist safety critical environments, such as warehouses, where hearing protection is enforced, making it more difficult for workers to identify and localise sound and reducing their situational awareness (Smalt et al. 2020). Noise-cancelling headphone users also lose aural information about their surroundings, causing them to lose awareness of their surroundings (Shrivastava et al. 2025). From these examples it can be seen that when hearing is impaired situationally or otherwise, situational awareness is reduced, this can impact individuals' safety as their ability to detect and react to warning signals such as alarms or vehicle horns, is reduced. (Shrivastava et al. 2025; Fostick and Fink 2021).

In order to mitigate the impact impaired hearing may have on situational awareness, this project explores how sound can be visualised using augmented reality to aid the detection and localisation of sound, retrieving auditory information about one's environment that may have been lost.

A wearable prototype capable of detecting and visualising spatial sound was created using a microphone array and mixed reality headset. This prototype provides users with visualisations of surrounding aural information enabling them to locate and navigate towards sounds in their environment.

The prototype created uses a miniDSP UMA-8 microphone array (miniDSP Ltd. 2018), used to detect and localise sound. The microphone array consists of multiple microphones placed on a circuit board, the known distance between each microphone can be used to determine the angle of arriving sound, in this project this is done using an implementation of the SPR-PHAT algorithm (Desai and Mehendale 2022). The microphone array is mounted upon a Meta Quest 3 mixed reality headset (Meta 2023) which is used to visualise the direction and approximate distance of detected sounds over the user's vision. A total of three visualisation techniques were created, derived from techniques used in augmented reality navigation solutions and out of view object visualisation. Augmented reality (AR) navigation involves using visualisation techniques to help guide individuals towards destinations (LYNAM et al. 2024). On the other hand, Out of view object visualisation techniques are used to signal objects that exist outside of an individuals field of view.

Visualisations were derived from techniques used in AR navigation and out of view object visualisations because there exist no standard visualisations for conveying spatial sound using augmented reality. The three visualisations created differ in terms of the explicitness of their cues and how they are positioned relative to the user.

Informed by existing AR navigation visualisation solutions and out of view object visualisations designs, we aim to discover what visualisation techniques are most effective when used to convey

spatial sound. We also aim to explore how individual and pair combinations of visualisation techniques compare in effectiveness at localising sound, measured through performance metrics as well as survey and interview responses.

To assess the effectiveness of each visualisation technique created an evaluation was conducted involving participants using each visualisation technique and pair combination to localise spatial sounds in a simulated environment. Performance metrics such as distance to target and angular error were measured to help understand each visualisations effectiveness at aiding the localisation of spatial sounds. Participant surveys were also used to gather user preferences and measure the perceived workload and visual demand of each visualisation. Participant interviews were also conducted to gain better insight into the reasoning of participant survey responses and gain understanding of their experience with each visualisation and the system as a whole.

It was found from the evaluation that user preference was varied with at least one participant regarding each of the 3 visualisations or 3 pair combinations as their most preferred. It also became evident that it is important for visualisations of spatial sound to preserve the omnidirectional nature of hearing. Pair combinations of visualisations proved to be effective at aiding in sound source localisation, with one pair being most preferred by the highest proportion of users. In addition, it was found that participants preferred using different visualisations at different distances from the sound source, suggesting flexible and adaptable visualisations should be created.

Overall, this project makes two contributions. Firstly, presenting a prototype system which visualises the location of spatial sounds using a mixed reality headset and a microphone array to sense the direction of spatial sounds. Secondly, by evaluating the design of spatial feedback for such sensing in a simulated VR environment, we demonstrate how different designs support user performance, confidence, and usability in different ways. We also propose beginning design guidelines for visualising aural environments.

2 | Background

In this chapter, we will explore the existing work done in sound localisation as well as other applicable fields of research such as navigation in augmented reality and visualising out of view objects. We will also discuss what has been achieved so far using wearable microphone arrays.

2.1 Microphone Sound Source Localisation

To visualise one's aural environment we need to be able to detect and measure the location of spatial sounds. Humans use aural information from two ears to localise sound, our system will utilise a microphone array to do so.

Microphone arrays have been used to localise sound sources in a wide range of scenarios such as robot audition, which enables robots to understand and react to surrounding sounds, drone audition, and autonomous vehicles (Löllmann et al. 2017; Marques et al. 2022; Wakabayashi et al. 2020).

When localising sounds using a microphone array, it is difficult to reach the same performance of the human ear using only two microphones, so it is common to use a microphone array that consists of several microphones (Desai and Mehendale 2022). Many algorithms such as steered-power beam-formers, subspace decomposition methods, and machine learning approaches, can be used to calculate the direction of arriving sound (Michaud et al. 2022). In this project, we make use of Time Difference of Arrival (TDOA) algorithms. As sound arrives at one microphone on the array it takes a certain amount of time for it to reach the next, using the difference in arrival times and the known distances between the microphones, TDOA algorithms can estimate the angle of the arriving sound. The degree of accuracy to which these algorithms can measure angles of arriving sound varies with microphone array geometry, the number of microphones, the distance of sound detected, the sample rate of measured sound, and reverberation present (Chung et al. 2022). In ideal conditions indoor benchmarking of proposed TDOA algorithms have shown angle error rates as low as 0.74 degrees (Chung et al. 2022).

Desai and Mehendale (2022) discuss how robotic ears can be developed to simulate the function of the human ear. To compare to the human ear's function, a robotic ear must be able to filter out noise and reverberations to detect sound, locate the sound, and classify what type of sound it is, for example if it is speech or an alarm. We focus on the ability to detect and locate a given, pre-identified sound in this project, we will not be classifying the detected sound.

Desai and Mehendale (2022)'s systematic review of sound source localisation systems found that most sound source localisation systems are only able to capture the azimuth (the horizontal) angle of arriving sound, with few able to estimate the distance and elevation of the sound source. Due to the limitations of the sound localisation algorithm used in this project, we will also only be able to capture the azimuth angle of arriving sound, we will estimate an approximate distance by calculating the root mean square (RMS) of the microphone array's audio signal. It is also noted in the review that reverberations negatively impact the performance of these systems creating uncertainty in the localisation systems accuracy. In this project, we benchmark the microphone array in a reverberant environment and visualise the error that occurs, displaying the capabilities of the sound source localisation system to the user.

2.2 Wearable Sound Source Localisation

Prior work exists in creating wearable systems that carry out sound source localisation, they commonly use a microphone array to detect spatial sounds as done in this project. Different types of feedback can be provided to the user to convey the location of sounds, we will discuss the use of haptic and visual feedback.

2.2.1 Haptic Solutions

Devices that use haptic feedback to convey the location of sound have been created for use by those with hearing impairments, aiming to help aid them in localising sound (Daoud et al. 2015; Michaud et al. 2022). The form factor of such devices is commonly a custom built haptic belt with motors placed along the perimeter of the belt, used to notify the user of the direction of detected sound.

Michaud et al. (2022) developed a haptic belt that was used to inform users of the direction of arrival of spatial sounds. The belt was equipped with eight microphones forming a microphone array that used to estimate the direction of arrival of sounds using algorithms based on GCC-PHAT, a TDOA method. The belt also used 15 haptic motors placed along the perimeter of the belt which were activated corresponding to the direction of arriving sound. The sound localisation method implemented was benchmarked on several different sound types such as truck horns, phone rings, and car horns. The results showed a mean average error of 2.90 degrees, displaying that wearable sound source localisation devices can provide accurate sound source localisation. The device in this paper was later evaluated to discern if the device aided in sound localisation for participants with simulated hearing impairments (Rousiouk et al. 2022). Results showed that the haptic belt only improved sound localisation of sounds in front of the user, with users rating the functionality of the device highly but rating the weight, appearance and comfort of the device lowly. These results suggest that although a haptic belt may provide accurate feedback and measurements of sound source locations, the physicality of the device hinders its adaptability by possible users due to its weight and appearance. For this reason, we have chosen to create a visual solution in this project.

2.2.2 Visual Solutions

Using visual feedback instead of haptic feedback may present a lighter, more user friendly device modality in which users may be able to use microphone arrays currently integrated in smart glasses to view visualisations of detected sound. For example, Lin et al. (2024) developed a beam-forming algorithm using the microphone array integrated into the Project Aria glasses (Engel et al. 2023). Beam-forming methods combine the audio signals from a microphone array to focus on sound arriving from a certain direction, filtering out incident sound arriving from other directions. The beam-forming algorithm was successful and was adapted for directional speech recognition. This displays the capabilities of current smart glasses technology, and as smart glasses become more widely adopted (*Smart Glasses Market Size, Share | Industry Report, 2033* 2026), they may offer a more familiar and socially acceptable format to assist sound localisation, rather than other wearable modalities such as haptic belts.

Although Lin et al. (2024)'s prototype was capable of performing directional speech recognition, it was not reported that it could accurately detect sounds from all surrounding angles of the user. However, wearable systems have been created that are capable of this. For example, Bosco and Tandi (2025) developed a system which integrates a microphone array into a wearable augmented reality headset. Their system reportedly performed well in realistic reverberant environments and was capable of localising sound in 3D; giving the azimuth, elevation and distance of spatial sounds (Bosco and Tandi 2025). Their algorithm uses Steered Response Power with Phase Transform (SPR-PHAT) as well as a Convolutional Recurrent Neural Network (CRNN) to

achieve a median localisation error of under 5 degrees for both the elevation and azimuth of spatial sounds, with the processing latency under 25 ms. This shows that microphone arrays can be implemented into current wearable technology to provide users with accurate measurements of the location of spatial sounds. What is lacking from this work is the ability to visualise the directional measurements of sound, something that this project will explore how to do effectively.

Although not visualising the direction of detected sounds, Asakura (2023) developed a system which presents classification labels of surrounding sounds in augmented reality, used to improve everyday tasks for deaf and hard of hearing people. They found that the respondents found everyday tasks more comfortable using the system. They also provide the spectrogram of sound being picked up by the microphone, suggesting that with training provided, users can better understand the sound in their environment using the spectrogram. This work showcases that visualisations of sound, in this case the classifications and spectrograms of sound, can prove beneficial to those with permanent hearing impairments. Comparably, this project aims to discover effective designs for visualising the location of sound, evaluating usability and user preference as a first step towards understanding how such visualisations might benefit users with impaired hearing.

Prototypes have also been created that display the characteristics of human speech in one's aural environment. Guo et al. (2020) created an augmented reality wearable prototype that visualised sound classification labels, live transcription of detected speech, and the detected location of sound. They used circular arcs in 12 discrete horizontal locations to present the angle of located sound stating that this visualisation technique was preliminary. As this visualisation technique of the location of sound was intended to provide lightweight awareness within a broader UI containing multiple aspects including classification and live transcriptions of sound, the arc-based approach was not adopted in the visualisations created in this project.

Live transcriptions have also been supplemented with the direction of arriving sounds to help aid those with hearing impairments contribute to group conversations (Dementyev et al. 2025; Jain et al. 2015). Dementyev et al. (2025) created a system which improved sound perception in group conversations using a mobile device. The sound localisation algorithm provided the direction of spatial sounds along the azimuth, similarly to the sound localisation algorithm used in this project. They integrated a microphone array into a custom built mobile phone case. The mobile phone screen presented the direction of each speaker as well as a live transcript of each speaker during the conversation. Through a user study they found that all users found the presentation of direction of arriving sound, in this case speakers, to be useful. This showcases how microphone arrays have been implemented into existing mobile/wearable hardware to help aid those with hearing impairments by visualising auditory information such as the direction of arriving sound and live transcripts of speakers in a group conversation.

Jain et al. (2015) also built a proof of concept device for visualising group conversations on a head-mounted display, exploring design preferences across eight different designs with deaf and hard of hearing participants. The proof of concept relied on a non-wearable microphone array limiting use to stationary group conversation. Their study focused on design preferences from static visualisations from which emerged a final chosen visualisation design. This design consisted of a top down view of a circle with arrows placed on the perimeter of the circle to convey the direction of arriving sounds, this inspired the design of the Radar visualisation later discussed in Chapter 3 and implemented in this project. This project also aims to evaluate the use of visualisation techniques by using them in a wearable context to localise sound in real time, something that this project contributes through controlled user study performance metrics such as distance to target and angular error.

2.3 Auditory to Visual Substitution

When auditory perception is impaired, one potential solution is sensory substitution, this is where information normally sensed through one sensory modality is represented through another. By translating auditory information into visual cues, as done in this project, some information about the surrounding environment can be regained (Wilson 2023). When substituting auditory information for visual feedback, it is important to understand the difference in how humans naturally perceive both sound and vision. Human vision is naturally unidirectional meaning vision is perceived from one direction and when an object moves outside of one's field of view they can no longer see it. Unlike vision, sound is omnidirectional meaning that one hears sounds from all directions regardless of where they are facing (Wilson 2023). A challenge that arises is that visualisations of aural environments must show the location of sound sources that are outside of the user's field of view, this issue is addressed in work done in visualising out of view objects in augmented reality.

2.4 Out of View Object Visualisations

As we are visualising sound using augmented reality devices, the user's field of view will be further constricted, making visualising objects outside this field of view even more common and important. There is existing literature focused on visualising out of view objects in augmented reality with many different visualisation techniques proposed (Quinn and Gabbard 2024).

Quinn and Gabbard (2024) conducted a systematic review of visualisations of out of view objects in augmented reality. They emphasise that there are no current design standards regarding how to design effective visualisations. In the review, to determine the effectiveness of each design, they compared 37 different visualisations based on location accuracy, task completion time, head movement, mental workload, usability/user preference. From this comparison and analysis of visualisations they were able to derive new design guidelines and highlight what properties of different visualisations are effective. Some example design guidelines proposed were; provide feedback to the user, encode direction and distance, use subtle cues instead of explicit cues when possible. These guidelines are also supported in the review by literature considering perceptual and cognitive processes, and aim to enable designers to create effective visualisations which require minimal visual attention.

EyeSee360 was the most effective visualisation found in the review overall for encoding both direction and distance. This visualisation involves an ellipse that represents a top down view of one's surroundings, placing markers corresponding to the position of surrounding objects (Gruenefeld et al. 2017).

Another visualisation that was reviewed was MonocularAR, one of two visualisations reviewed which used only the periphery of the user's vision to encode the direction of objects (Gruenefeld et al. 2018). They used real LED lights surrounding the lenses of a mixed reality headset to direct users towards objects outside of their field of view. They also developed a virtual on-screen version which simulated the light effects on the edge of the rendered display. The on screen method resulted in faster search times and was preferred by users in a user study.

The two visualisations mentioned above showcase different techniques, one showing a top down overview providing the explicit direction of out of view objects, and the other being ambient in nature only using the periphery vision of the user to indicate the direction of objects to the user. These have proved effective in the review and provide a solid foundation for creating effective visualisations of spatial sound which commonly exist outside of the user's field of view. Therefore, two visualisations created for this project are derived from EyeSee360 and MonocularAR respectively.

2.5 Augmented Reality Navigation

Augmented reality (AR) navigation involves overlaying information over user's field of view to create navigation solutions that point and guide user towards locations LYNAM et al. (2024). Many of these visualisations overlay directional cues to guide users, these visualisation techniques provide a foundation for creating visualisations that convey spatial sound as the directional cues can be used which encode the direction and approximate distance of sounds. These directional cues can cue the position of spatial sounds enabling the localisation and navigation towards sounds. AR navigation solutions are often evaluated by measuring performance metrics such as the distance from the target and the time taken to navigate towards the target, this approach was also used to evaluate the effectiveness of the visualisations created in this project (Vilar et al. 2014).

2.5.1 Auditory Solutions

There exist navigation techniques that do not use visual feedback as done when using augmented reality. One approach is using auditory cues to guide users instead. For example, Zhao et al. (2020) developed a navigation solution that uses smart glasses to guide users towards destinations using audio feedback played through the glasses speakers. Their evaluation of their solution found that it improved navigation for participants with visual impairments, however, participants with a full field of vision completed navigation tasks faster when using visual feedback. Zhao et al. (2020) found that all participants wanted to combine both audio and visual feedback to aid their navigation. This supports Qiu et al. (2025) who found in their review of the use of AR in way-finding that users consistently disfavoured systems which depended only on audio to provide feedback. This project does not explore auditory cues for conveying spatial sound, as Zhao et al. (2020) found visual feedback to be more effective for navigation performance, and auditory guidance may be less reliable when users hearing is impaired. Instead, this project focuses solely on visual methods for representing spatial sound.

2.5.2 Haptic Solutions

Similarly to how wearable sound source localisation systems exist that use haptic feedback to convey the location of sound, there exist navigation solutions which use haptic feedback to point and guide users towards destinations. For example Gay et al. (2020) created a prototype that uses a haptic belt to help guide users and assist independent navigation, their study was focused on deaf blind individuals and that found the prototype improved navigation in participants with deaf-blindness. Our project will not be accessible to such groups as it relies heavily on vision to present the direction of sound. LYNAM et al. (2024) also argue how emerging consumer AR devices such as the Apple Vision Pro reflect the growing adoption of mixed reality technology, whilst haptic feedback systems lacks the same attention and consumer attention.

Current wearable devices that substitute auditory information for haptic feedback have also been discussed in Section 2.2.1, finding that the appearance and comfort of these devices may hinder usability and the adoption of such technology in the future (Rousiouk et al. 2022). For these reasons, this project focuses only on using visual techniques to convey spatial sound.

2.5.3 Visual Solutions

We will now discuss visual solutions which use augmented reality to aid navigation. Qiu et al. (2025) conducted a systematic review of the use of augmented reality in human way-finding and found that visual guidance proved more effective than auditory guidance. They also found that overview maps of one's surroundings vastly improved performance and reduced mental workload in navigation tasks. Similarly to literature involving out of view visualisations, this review suggests that interfaces should be kept minimal in order to remain accessible to users with

cognitive impairments. They also argue that visualisation should be customisable and modular in nature where possible.

They highlight that there is no dominant visualisation that is being used in AR navigation solutions. They also state there is a lack of research on the effectiveness of these modalities as a whole. From evaluating literature they concluded that in terms of user experience, users preferred a simple streamlined visual presentation such as the standard arrow. It is also mentioned that users became frustrated when the localisation technique used by these modalities did not display an accurate representation of the real routes, this issue is considered in this project as the sensing of spatial sounds location is not fully accurate and has discrepancies to the real world location of sounds. Some papers reviewed by Qiu et al. (2025) displayed an interest in lowering the localisation error (Gang and Pyun 2019), however none of the papers developed methods of visualising this error. As previously discussed, this project will convey this error in the visualisations implemented in this project to minimise possible user frustration caused by inaccurate representations of spatial sound.

LYNAM et al. (2024) surveyed AR navigation solutions and developed a framework that can be used to categorize and understand the solutions surveyed. The survey provides an overview of the current state of augmented reality solutions and challenges faced. They surveyed 71 papers and found that many overlooked the visualisation aspect of their solution. The visualisation aspect of a system impacts adoption of AR tech by the public so it is important to create effective visualisations that are preferred and aid users in navigation. The papers surveyed also only evaluated the effectiveness of localisation using one visualisation and never evaluated differences between visualisations. The papers surveyed use varying localisation techniques such as direct-sensing and triangulation, however, the sound source localisation technique presented in this project is not covered in any current AR navigation literature. The survey also found that visual feedback proved to aid navigation the best in comparison with verbal and haptic feedback.

The most common visualisation LYNAM et al. (2024) found is the arrow, with it being implemented in various ways. To assess the effectiveness and suitability of using this common AR navigation visualisation for localising sound, it will be implemented as one of the visualisation in this project. Its effectiveness will be compared with egocentric visualisations as well as combinations of visualisations. In many of the papers visualisations are combined, often visualisations that appear relative the position of the users body, such as arrows, are paired with head-mounted visualisations, such as a map, which appear statically in a section of the user's vision providing a top down overview of the environment.

The survey ultimately found that combinations of visualisations proved beneficial for navigation, suggesting that this was due to users being able to fall back to the second visualisation if they failed to interpret meaningful information from the first. They do highlight the risk that multiple visualisations may result in information overload, garnering too much visual attention. This recommendation of multiple visualisations supports the recommendations made in Quinn and Gabbard (2024)'s review of out of view object visualisations, which proposes that combinations of visualisations must avoid causing visual clutter in order to prove effective. This project will investigate if combinations of visualisations also proves effective when visualising spatial sound.

2.6 Summary and Research Gap

Existing research demonstrates that microphone arrays can be used to create wearable sound source localisation systems that gather information about users surrounding spatial sounds, such as direction for example. Whilst wearable solutions have been developed that visualise the direction of sound using augmented reality, many of the visualisations are preliminary and there is limited work evaluating the effectiveness of the visualisation techniques used. In light of the above there is limited understanding of how spatial sounds can be effectively represented visually in augmented

reality, this research aims to fill this gap providing some insight into what constitutes an effective visualisation for representing auditory information

Due to hearing being omnidirectional, out of view object visualisations provide a foundation to create visualisations which display spatial sounds that are located out with the users field of view. Visualisation techniques used in AR navigation also provide insight into how to design visualisations that effectively use directional cues to guide users towards locations, in this project we will use directional cues to point towards the sensed location of spatial sounds.

It is also evident that combinations of visualisation techniques have also been found useful in both AR navigation and out of view object visualisations, this project aims to explore if combinations also prove effective for visualising spatial sounds. By evaluating the effectiveness of visualisations derived from AR navigation and out of view object visualisation, we aim to find what constitutes an effective visualisation and propose initial design guidelines for visualising spatial sound, derived from the findings of our evaluation.

3 | Design

3.1 Overview

Derived from differing visualisation techniques found from the literature discussed in Chapter 2, three visualisations were created. The three resulting visualisations are Arrow, Radar, and Lights. Aligning with the design guidelines proposed by Quinn and Gabbard (2024) for visualising out of view objects, each visualisation encodes the distance and direction of sound. The distance measured is approximate as the sensing of distance of sound is limited when solely using a microphone array (Desai and Mehendale 2022). The uncertainty of the prototype's sensing capabilities were also encoded into each of the visualisations.

3.2 Visualisations Created

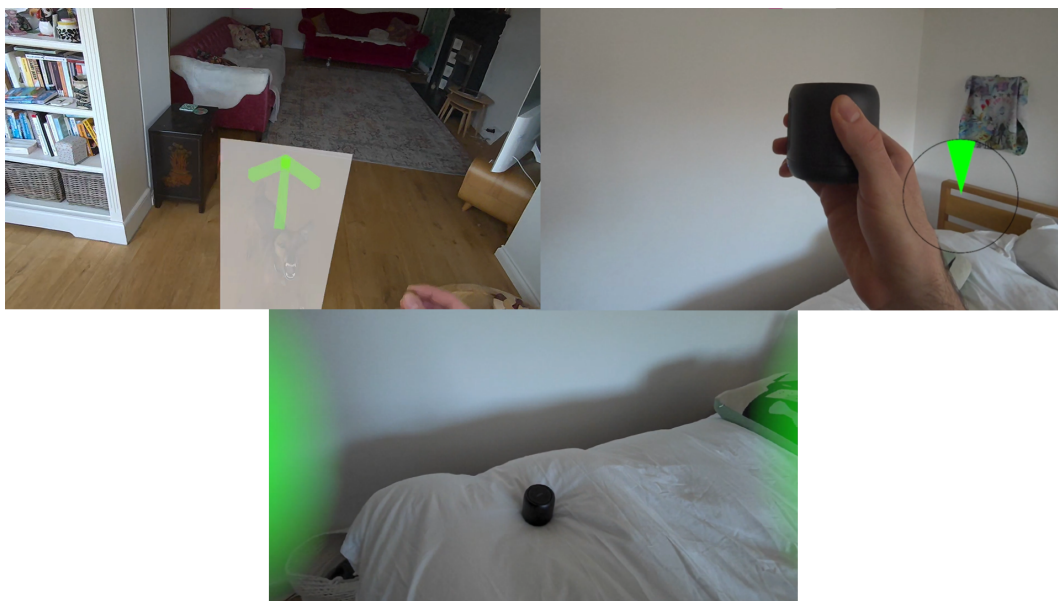


Figure 3.1: Images of visualisations designed for this project. Top left: Arrow visualisation detecting sound from dog in front of user. Top right: Radar visualisation detecting sound from speaker in front of user. Bottom middle: Lights visualisation detecting sound from speaker in front of user

All visualisations created use the egocentric frame of reference, meaning they are placed relative to the users position. The other possible frame of reference is exocentric, where visualisations are placed relative to the user's wider environment (Markov-Vetter et al. 2020). Egocentric visualisations were chosen because Markov-Vetter et al. (2020) found them to be more effective than exocentric visualisations at navigating users towards off-screen targets.

3.2.1 Arrow

The Arrow was chosen because it is the most common augmented reality visualisation technique found in AR navigation and regularly requires low cognitive effort to interpret (LYNAM et al. 2024). Its simplicity makes it a strong baseline against the other two visualisations, Radar and Lights, chosen which are more complex possibly requiring more cognitive load to translate the direction presented to the actual direction between the user and the sound source. The Arrow is also visually placed in the users surrounding environment, pointing towards the real direction of detected sound, unlike the other two visualisations created which are head-mounted meaning they are anchored to the position and rotation of the users head, and therefore remain in a fixed area of vision.

The Arrow's design was derived from other arrow visualisation techniques used in the papers surveyed in LYNAM et al. (2024)'s survey of augmented reality navigation (Yunardi et al. 2022; Matuszka et al. 2013). Differing from the other visualisations created in this project, the Arrow is not head-mounted and does not appear in one's vision if a sound is located outwith the users field of vision. By evaluating the effectiveness of the Arrow in comparison with the other two visualisations, we aim to discover how visualisations that are visually placed in the user's surrounding environment and are anchored to the user's body, compare to head-mounted visualisations that are visible at all times.

The arrow visualisation was adapted, with a cone placed around the Arrow to display the systems localisation uncertainty. The opacity of both the uncertainty cone, and the Arrow, encodes the approximate distance of the sound source, this was chosen as the size of the visualisation is already used to encode the uncertainty of the sound source localisation technique.

3.2.2 Radar

The Radar is derived from Gruenefeld et al. (2017)'s EyeSee360 visualisation technique, the most effective visualisation found in Quinn and Gabbard (2024)'s systematic review of AR visualisations of out of view objects. The EyeSee360 technique is a head-mounted visualisation technique that places markers, representing objects, on an ellipse. The visualisation provides a top down overview of a user's surrounding environment allowing them to visualise the horizontal and vertical direction and distance of objects outside of their field of view. Due to the sensing capabilities of the prototype implemented in this project, we are only able to visualise three different possible values of approximate distance, the azimuth of one detected sound source at a time. Because of this limitation the EyeSee360 visualisation was adapted to better portray the restrictions of our prototypes sound source localisation capabilities. The Radar visualisation implemented is also inspired by the visualisation used in Jain et al. (2015)'s proof of concept, which presents a top down view of surrounding sounds, which includes a circle with arrows lying on its circumference indicating the direction of sound.

This design was chosen because it explicitly shows the angle of the detected sound similar to the Arrow but remains visible if the sound detected is out of view, due to the visualisation being head-mounted. As the visualisation is anchored to the users view, this allows for a top down view of the surrounding environment to be envisioned. This is similar to how LYNAM et al. (2024) describes how maps can be paired with other visualisation techniques to provide a broader overview of the environment aiding in navigation. By evaluating this technique against others in our evaluation, we aim to discover if this top down overview approach is effective at aiding users locate spatial sounds in their environment.

The Radar visualisation implemented consists of a notch placed within a circle, where the circle represents a top-down overview of the space surrounding the user. The notch begins in the centre of the circle and points outwards towards the perimeter of the circle, the angle at which this notch is pointed indicates the azimuth of detected sound relative to the user's current orientation.

For example, when sound is detected in front of the user the notch points towards to top of the circle, while sounds detected behind cause the notch to point to the bottom. The width of the notch correlates with the uncertainty of sound source localisation technique. The opacity of the notch encodes the three possible approximate distances of the detected sound source, similar to the Arrow this was chosen as the size of the notch correlates with the uncertainty of the sound source localisation technique.

3.2.3 Lights

The Lights visualisation is derived from the out of view objects visualisation technique; Monocular (Gruenefeld et al. 2018). Monocular uses simulated lights on the periphery of the user's vision to direct them to objects outside of their view. This visualisation technique was chosen because it is only displayed on the periphery of the user's vision, potentially requiring less visual demand than other more explicit visualisations such as the Arrow. This aligns with Quinn and Gabbard (2024)'s design guideline for out of view object visualisations, which stated that subtle cues should be used instead of explicit cues where possible. They claim that subtle cues are not likely to affect search performance, by implementing a visualisation that uses subtle cues we will see through our evaluation if this design guideline holds true for visualising spatial sound. The visualisation technique was adapted to only encode the azimuth as unlike Monocular our system only detects sources on the azimuth and does not display the elevation of sound.

This visualisation encodes three directional states using two images placed on the periphery of the user's vision to simulate lights, one light is placed on the left of the user's vision and the other on the right. If the user is facing within 30 degrees of the sound detected, corresponding to the uncertainty of the systems sound source localisation method, both lights are displayed, this visualisation does not present the precise angle detected, and instead displays the uncertainty range where the sound source is detected. To represent the second state where sound is closest to the left of the user, the left light is displayed. Likewise for the third state where the detected sound is closest to the right of the user, the right light is displayed. The lights shown increase in opacity as the azimuth of the sound detected becomes smaller, correlating with the detected sound being closer to the centre of the users field of view. This change in opacity provides subtle feedback corresponding to the azimuth of the detected sound. The approximate distance of the sound detected is encoded through the height of the lights, the taller the lights the closer the measured approximate distance to the sound source is.

3.2.4 Smoothing

To improve the usability of the Arrow and Radar visualisations, linear smoothing was applied to rotation of the visualisations to reduce abrupt movement and prevent them from appearing overly unstable. This also improved visual continuity by ensuring that, in the case of the Arrow, the visualisation did not rotate large amounts instantaneously when the detected sound direction changed significantly between updates.

3.2.5 Combinations

All possible pair combinations of the Arrow, Radar, and Lights visualisations were created to evaluate the effectiveness of using combinations of visualisation techniques. Work in AR navigation solutions suggests combinations of visualisation techniques improve usability because if one visualisation fails to give the user effective feedback, the user can fall back to the second visualisation to aid navigation (LYNAM et al. 2024). By evaluating the combinations along with individual visualisations we wish to explore if combinations also prove effective for visualisation the location of spatial sounds.

3.2.6 Limitations

We chose to focus only on encoding the direction and approximate distance of spatial sounds. To provide more complexity, the Classification of sounds could have been attempted and visualised as done in existing work in wearable sound source localisation systems (Guo et al. 2020; Shen et al. 2012). However, the added complexity of visualising classifications would make it more difficult to discern and measure what aspects of each visualisation contribute to effective localisation of sound.

4 | Implementation

This project presents a wearable prototype system which visualises the location of spatial sounds using a laptop, microphone array, and mixed reality headset. This chapter will detail how the prototype system is implemented as well as benchmarked in order to present its sensing capabilities.

4.1 Microphone Array

The microphone array used in this project is the UMA-8 Microphone Array by miniDSP, a circular microphone array with an on-board Digital Signal Processor (DSP) miniDSP Ltd. (2018). The UMA-8's DSP provides an Direction of Arrival (DOA) method which calculates the direction of arrival of one sound source, the angle of this sound source can be accessed through the HID interface.

4.1.1 Sound Source Localisation Algorithm

The onboard DOA method of the UMA-8 provides the estimated angle of detected sound, however it is optimised for detecting and processing speech as a far field microphone limiting its ability to detect accurately sounds other than speech (miniDSP Ltd. 2018). It also does not give access to each individual microphone, making it difficult to estimate the approximate distance measurement needed in this project to convey the location of spatial sounds. Due to these limitations, we created our own DOA method using the UMA-8's RAW audio mode, this mode gives access to the raw microphone signal from each of the 8 microphones allowing custom DOA methods to be implemented (miniDSP Ltd. 2018).

To implement a DOA method, we created a sound source localisation algorithm using python, the method uses the raw microphone audio signals of the microphone array to estimate the direction of arrival of sound sources. Using the Pyroomacoustics library the sound source localisation algorithm implements a version of SRP-PHAT, a classic TDOA method, this method was chosen as it performs well in noisy and reverberant environments making the prototype more suitable for use in environments with poor acoustics (Velasco et al. 2016; Scheibler et al. 2017).

To approximate the distance from the detected sound source, the root mean square (RMS) of the microphone array's audio signal is computed. The RMS is then normalised between 0 and 1 and binned into three values representing approximate distance, far (≤ 0.2), medium (≤ 0.7) and near (≤ 1).

The onboard DOA method of the UMA-8 does provide a beam-formed audio signal of speech detected, this can be used to classify sounds especially speech.

An audio classification model that classifies the stream of audio outputted by the UMA-8's onboard DOA method was also implemented using Google's Media-Pipe Audio Classifier to classify audio into defined categories in real time (Google 2026). Although attempted, classification of the audio processed by our own DOA method was not achieved, subsequently the classifications of detected sounds were not used in the prototype with no visualisations developed in this project being adapted to display to the classifications of sound.

Despite not providing beam-formed audio, the SPR-PHAT algorithm we implemented detected all types of sound more accurately, as seen in benchmarking discussed further on in this chapter, and so was chosen as the DOA method to be used in the prototype.

Our final algorithm detects only one sound source at a given time but detecting more than one sound source is possible. For example, Grondin et al. (2022) developed sound source localisation techniques using a circular microphone array to detect up to 3 sound sources. They were also able to beam-form the audio from each sound source providing focussed audio from each source. We attempted to modify the algorithm used in this project to detect more than one sound source, however from trivial testing it was found that it was not able to consistently detect multiple sound sources at once and could not be used in the prototype. Due to this limitation, the visualisations implemented in this project are only able to visualise the location of one sound source.

4.2 Microphone Array Benchmarking

Because this project simulates use of the prototype and also visualises the uncertainty of the prototype's sensing capabilities, the sound source localisation technique was benchmarked in an untreated reverberant indoor environment enabling us to find the prototypes sensing capabilities and replicate them. To find the uncertainty of the different possible sound source localisation approaches available for the prototype to use, the onboard DOA method of the UMA-8 was benchmarked along with the created DOA method utilising an implementation of the SRP-PHAT algorithm.

A speaker was placed one metre away from the centre of the microphone array, it was moved at 30 degree intervals around the microphone array, shown in Figure B.1 in the appendix. This was repeated 3 times for both an alarm sound and with speech. The absolute difference between the real angle of the speaker and the measured angle was recorded and the mean and standard deviation of differences were calculated. To be conservative, we chose to display the largest average uncertainty reported from the benchmarking across both DOA methods which was 30 degrees.

To find the moving uncertainty, meaning how much the measured angle varied whilst detecting a stationary sound source, the microphone array was placed one metre away from the speaker, with the speaker placed directly in front of the microphone array so that the real angle between the two was zero degrees. The measured angle was recorded for 10 seconds to measure the deviation in measured angle. The highest standard deviation and range recorded was 1.25 and 8.00 respectively. To be conservative, a standard deviation of 2.00 and range of 10.00 was chosen.

4.3 Data Pipeline Overview

A laptop is worn in a backpack by the user, connected to the microphone array via USB. The laptop runs the implemented DOA method python script which measures the angle and approximate distance of arriving sound using the raw audio signals from the microphone array. The DOA method script also runs a web-socket server which broadcasts the angle and distance in a JSON message, it sends this message to clients every 0.05 seconds giving a send rate of 20 Hz. By connecting to the same network to which the laptop is connected, the headset connects to the web-socket as a client. The headset is then able to retrieve the message broadcast by the laptop and extract the approximate distance and angle measured.

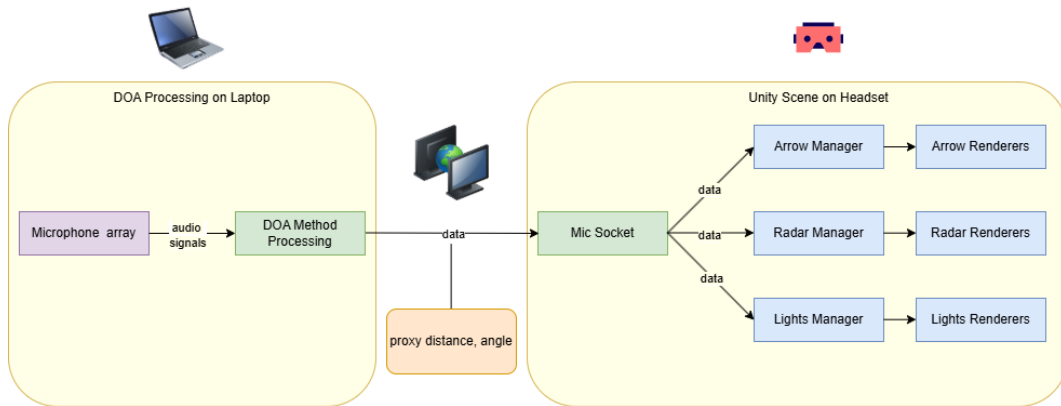


Figure 4.1: A diagram of the data pipeline of the system, detailing how data is processed on a laptop, sent to the headset and used to manage visualisations in the Unity scene.

4.4 Wearable Prototype Design

To overlay visualisations on top of user's vision the Meta Quest 3 mixed reality headset was used. The microphone array is mounted onto the top of the headset worn by the user with tape to keep it secure, as shown in Figure 4.2. The user also wears a backpack containing a laptop that is connected to the microphone array, the laptop sends a message containing the angle and proxy distance to the headset via the web-socket as described in the data pipeline over view above. The augmented reality environment that was built to the headset for the prototype was developed in the 3D game engine Unity (Unity 2026), using the Meta pass-through API (Meta 2025). This allows the visualisations to appear placed upon one's vision of reality seen through the cameras of the headset.

The Unity scene includes an object created called a mic socket which connects to the web-socket server, broadcasting the angle and approximate distance received from the connection to the scripts used for each visualisation. Using the approximate distance and angle received from the mic socket in the scene, visualisation scripts update 3D game objects and 2D images to materialise the design of each visualisation specified in Chapter 3. The mic socket implements a general mic socket interface, this allows the visualisations to remain the same between different simulated and non simulated mic sockets.



Figure 4.2: A photo of a user using the wearable prototype - a mixed reality headset with a microphone array mounted on top connected to a laptop contained in a backpack worn by the user

5 | Evaluation

In this chapter, we detail the design of the evaluation used to answer the research questions asked in this chapter. We then present the results found from the evaluation and discuss how they answer the research questions and evaluate our findings in the context of relevant literature. Finally, we propose initial design guidelines for creating effective visualisations of spatial sound using augmented reality.

5.1 Experimental Design

To assess and compare the effectiveness of the visualisation techniques created in this project, we conducted a controlled within-subjects study in simulated Augmented Reality using the Meta Quest 3 mixed reality headset.

Our study had one independent variable, which was the visualisation technique used. This independent variable had six conditions, three being the individual visualisations, and three being the pair combinations of these three visualisations.

Participants were tasked to locate sound sources in a virtual environment using each of the conditions. The simulated virtual environment for the study used an open source 3D model of a warehouse sourced from 3dwarehouse.sketchup.com (Mahaveer 2021). Sound sources were placed in this environment with all sound sources emitting an alarm sound at the same loudness, with one active at a time.

A virtual environment was chosen, instead of using the prototype in a real world environment, because it allowed access to a larger space and let us record measures, such as distance to target, accurately and efficiently through direct logging on the headset. The large open space meant sound sources and spawn points could be placed at varying distances from one another, enabling the effectiveness of each visualisation to be assessed across near, medium, and far distances. Short walls were placed at intervals throughout the warehouse to disrupt the user's path to sound sources, better simulating navigational challenges faced in real world environments.

A limitation introduced by using a virtual environment is that the prototypes sound source localisation technique is not being used, meaning results may not fully reflect real world performance, to help remedy this we simulated the prototypes sensing capabilities. To simulate the use of the prototype system, a virtual mic socket in the scene computes the angle between the user and the sound sources in the scene. Gaussian noise corresponding to the moving uncertainty found from the benchmarking of the microphone array is added to this angle to simulate the sensing capabilities of the microphone array.

In the wearable prototype, approximate distance is measured by computing the root mean square (RMS) of the microphone array's audio signal, which is then normalised between 0 and 1 and binned into three categories; near, medium, and far. To simulate the computation of approximate distance in the wearable prototype, we computed the distance between the user and the sound source with minimum and maximum distances to mimic the maximum and minimum thresholds used in computing the RMS for the approximate distance in the wearable prototype. This distance is then normalised between 0 and 1 and binned into the same three categories as done

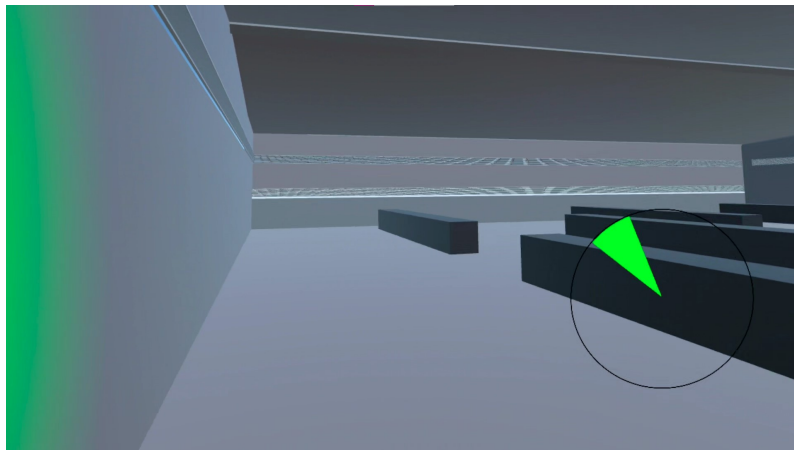


Figure 5.1: Screenshot of the virtual environment implemented can be seen with the Radar and Lights visualisation being used.

in the prototype. As all sound sources in the virtual environment emit the same alarm sound at the same loudness, distance was the only factor influencing the perceived loudness making this representation of distance reasonable. Noise was added to this distance calculation to simulate the variability found in the real system's RMS measurement. A limitation of this approach is that the amount of noise added was chosen trivially instead of being derived from benchmarking of the actual distance proxy measurement, meaning the simulated distance uncertainty may not fully reflect the real system's capabilities.

To assess the effectiveness of each visualisation technique performance metrics including distance to target, angular error, and response time were measured to help understand each visualisations effectiveness at aiding the localisation of spatial sounds. Participant surveys were also used to gather user preferences and measure the perceived workload and visual demand of each visualisation. Participant interviews were also conducted to gain better insight into the reasoning of participant survey responses and gain understanding of their experience with each visualisation and the system as a whole.

5.2 Research Questions

This study was designed to answer the following research questions:

- Informed by existing AR navigation and out of view visualisation designs, what visualisations are most effective for conveying spatial sound? (RQ1)
- Are combinations of visualisations effective for conveying spatial sound? (RQ2)

5.3 Procedure

Before participants began the experiment, they were given a brief demo of the functioning wearable prototype, allowing them to understand how each visualisation technique functioned whilst detecting sounds in their real aural environment. This also enabled participants to gain a better understanding of the systems possible applications.

A total of ten sound sources were placed randomly within the environment along with six spawn points also randomly placed. These spawn points were where the participant begun the trial. The trials of the experiment iterated through all sixty possible pair permutations of sound sources and spawn points for each participant, the order being random between every participant.

There were ten trials per each of the six conditions, giving a total of sixty trials. To give participants time to understand the individual visualisations themselves and not be overwhelmed by the pair combinations, the first three conditions of the study for every participant were the individual visualisation techniques. The order of these first three conditions was counter balanced using a balanced Latin square, this ensured learning effects between each condition was minimised. The last three conditions were the pair combinations of the visualisation techniques, the order of which was also counter balanced using a balanced Latin square to minimise learning effects.

Participants were prompted to attempt to get as close as possible to the sound sources and face the direction in which they believed the source to be in. Once they believed they had located the sound source they clicked a button on the headset's controller, logging their distance from the sound source, the angular error, and the time taken to complete the trial. Angular error refers to the difference in degrees between the direction of the participants vision on the azimuth and the location of the sound source on the azimuth. After the button was pressed participants began the next trial, with the next permutation of spawn point and sound source.

Because the distances between spawn points and sound sources vary, the time taken for each trial is not a reliable indicator of effectiveness. However, the shortest route between each spawn point and sound source could be calculated and the time taken to follow this shortest route could be calculated and compared to the time recorded to complete the trial. This was not done in this project but it would give a balanced measure of time across each trial and allow each visualisation to be compared in terms of this balanced time taken metric.

Participants were given time before the experiment to practice using each visualisation to locate sound sources in the virtual environment. Once they felt comfortable using the visualisations, they began the first condition.

5.3.1 Surveys

After each condition was complete, participants removed the headset and completed a survey regarding the visualisation technique just used.

To measure the amount of cyber sickness experienced by participants after each condition, the first question in the survey was the Misery Scale (MISC) (Bos et al. 2005). This questionnaire measures the progression of motion sickness, if the participants responded higher than three, indicating beginning symptoms of nausea, they were instructed to take a break and only allowed to continue the experiment once they did not feel nauseous and reported a response to the misery scale of three or below. This ensured that all participants were physically well and that cybersickness did not affect participants performance or survey response during the study.

To allow us to compare the perceived workload between each visualisation, participants reported workload using the NASA-TLX questionnaire across six dimensions on a 21-point scale (Hart and Staveland 1988).

To evaluate visual workload and attentional impact of each visualisation technique, a questionnaire of three 7-point Likert items was created. This questionnaire measured the visual demand of each visualisation, how much its visual information overwhelmed the user, and how much the visualisation interfered with awareness of their surrounding environment.

User confidence when using each visualisation to localise sound was also measured through a single item 7-point Likert question.

After all conditions were complete, the participants completed a survey regarding all conditions. To measure user preferences, participants were asked to rank the visualisations from best to worst. They were then asked to select the visualisations, or combination of, they would prefer to use for near, medium, and far distances from sound sources. This assessed if participants preferred different visualisations for different ranges of distance, suggesting if participants were likely

to swap between visualisations in real use of the system indicating if flexible visualisations are required.

The participants were then asked to complete a questionnaire gathering the perceived usability of the system in four different scenarios, and if users were likely to use the system themselves in these scenarios. To judge how appropriate the system was in different contexts where hearing is impaired along with varying safety risks involved, the four scenarios chosen involve using the system being used whilst hearing is impaired in contexts with differing levels of safety concerns.

5.3.2 Participant Interviews

After participants had completed the final survey, they were asked nine questions in a short semi-structured interview.

The first three questions gathered insights into their survey responses, gaining understanding of their reasoning and thought process for their ranking of visualisations and chosen visualisations for each distance.

The next three questions and the final question, assessed usability of the system, asking users if they would use the system how they would use it in their everyday lives.

The seventh question asked what situations the participant thought use of the system would be inappropriate or detrimental. This helped us to understand the perceived limitations of the system and also understand the possible detrimental effects the system may have.

The eighth question gathered any suggested improvements that participants had for the system as a whole or particular visualisations. This helped motivate further work as well as understand what aspects of the visualisation designs participants thought could be improved to provide more effective representations of aural environments.

5.3.3 Limitations

The experimental design listed above allowed us to gather meaningful results to answer the research questions defined as well as gain further understanding to the impact of our design decisions. However, there are limitations to the evaluation that impact the generalisability of the results gathered.

To begin with, all visualisations could have been counterbalanced together instead of being split into the individual and pair combinations as done in our experiment. This would allow us to better assess the comparison in effectiveness between single visualisations and pair combinations of visualisations. We counter balanced the individual and pair combinations separately to ensure that participants understood how each individual visualisation technique functioned, and were comfortable using them before they were trailed with using the pair combinations. This issue of limited practice could have been mitigated by having participants practice with each condition for longer, ensuring that they were comfortable using the pair combinations before the experiment began.

Another limitation of the experimental design is that the usability of each visualisation technique was not adequately measured. To better assess the usability of each visualisation technique, a usability questionnaire such as the System usability scale (SUS) (Brooke et al. 1996) could have been included in the survey taken after each condition. Due to this limitation in this experiment we rely on discussion from the semi-structured interviews and participant rankings of all visualisations from surveys to analyse the usability of each visualisation.

In addition to the other limitations, the sample of participants used in our study limits the relevance of the results found from the study. The gathered sample of participants was made up mostly of individuals aged 18-24, this limits the generalisability of our results to other age ranges, especially

Table 5.1: *Distance to Target summary statistics by visualisation*

Visualisation	n	Mean(m)	Median(m)	SD(m)	Min(m)	Max(m)	Range(m)
Arrow	81	1.42	0.78	1.85	0.55	11.55	11.0
Radar	81	1.71	0.8	3.37	0.18	25.09	24.91
Lights	81	3.99	1.86	5.39	0.59	27.81	27.22
Arrow & Radar	81	2.44	0.78	4.93	0.25	25.55	25.3
Arrow & Lights	81	1.96	0.87	2.88	0.39	17.02	16.63
Lights & Radar	81	2.2	0.94	3.5	0.13	17.39	17.27

given many target users who experience hearing impairments may include older adults (Löhler et al. 2019). To remedy this, a more diverse and representative sample should be chosen in future work.

A further limitation of the gathered sample of participants is that the sample size was small with only nine participants, this still allowed us to gather enough data to derive meaningful findings from however, a larger sample size would improve the generalisability of our results and help to obtain statistically significant results.

5.4 Results

For the study 9 volunteer participants (8 male, 1 non-binary) were collected from a convenience sample, 8 participants were aged between 18 and 24 years, and 1 participant was aged between 55 and 65 years. All participants reported having no colour vision impairments, seven participants had normal vision, and 2 participants had normal-corrected vision.

5.4.1 Performance Metrics

To evaluate the effectiveness of each visualisation technique at aiding participants locate spatial sounds, the distribution of performance metrics for each visualisation is analysed. We also plot angular error against distance to target to understand the relationship between them. We will not assess the response time metric as the distance between sound sources and spawn points differed as previously discussed in Chapter 5.

To minimise the effect of learning effects and control for initial adjustment, the first trial for each condition was excluded from the analysis of performance metrics.

Distance to Target Distribution

Figure 5.2 shows a violin box plot of distance to target distribution by visualisation. Distance to target refers to how close participants were able to get to the sound sources they were locating in the experiment.

It is clear that the Lights visualisation has the largest spread of values obtaining the largest interquartile range, and the highest median amongst the visualisations. This is confirmed from Table 5.1 where Lights has the highest median of 1.86 metres and largest standard deviation of 5.39 metres. This suggests that the Lights visualisation provided the least effective feedback in relation to the distance from sound sources. Lights was the only visualisation techniques evaluated that encoded the approximate distance using the size of the visualisation instead of the opacity. This suggests that encoding approximate distance using the opacity of the visualisation is more effective and the distance perceived by a change in opacity is more proportional to the actual increase of distance compared to the distance perceived by an change of height.

Arrow had the lowest mean of 1.42 metres, the lowest median of 0.78 metres, and the lowest standard deviation of 1.85 metres, shown in Table 5.1. This suggests that the Arrow visualisation

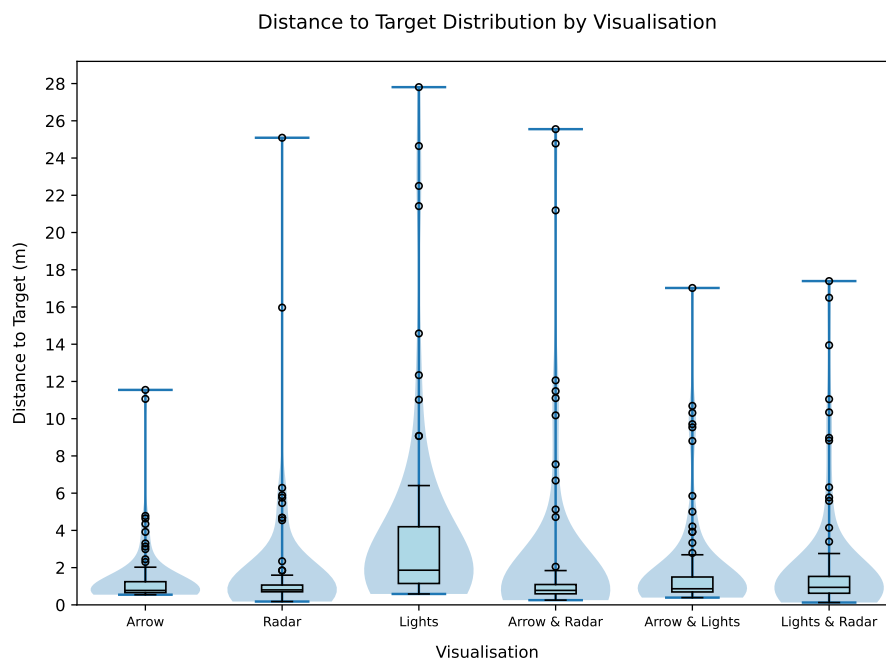


Figure 5.2: Violin box plot of distance to target distribution by visualisation. Lights has the largest spread and highest median. The other visualisations all have smaller spreads and lower medians.

provided participants with the most effective distance feedback enabling them to perform consistently among trials, resulting in the lowest median and standard deviation. However, the Arrow does not have the lowest minimum distance to target measured, the Arrow had a minimum of 0.55 metres whilst Lights and Radar has the lowest minimum of 0.13 metres followed by Radar with 0.18 metres, as seen in Table 5.1 .

All three of the lowest distance scores include the Radar in the visualisation technique, implying that users were able to get closest to the sound using the Radar than they would be able to using the Arrow or the Lights. This could be due to the fact that unlike the arrow, the Radar is not placed in the users surrounding environment and is instead implemented as a heads up display that is mounted to the users head position. As the Arrow is visually placed in the users surrounding environment, it points to a direction in the environment and ends a metre away from the users head position. The Arrow may suggest to the user that the sound source is positioned near its end, making it difficult for users to interpret that sound sources are located close to where the Arrow begins.

We can see also from Figure 5.2 and Table 5.1, that the combinations of visualisations perform comparably to the Arrow and Lights visualisations, portraying similar medians and spread of values. This suggests that using combinations of visualisations does not negatively impact the users ability to get close to sound sources, however the results do not suggest that they outperform the individual visualisations of Arrow and Radar, in terms of enabling users to accurately depict the location of sound sources.

Angular error distribution

Figure 5.3 shows a violin box plot of the angular error distribution by visualisation. This displays how close users were able to face the location of the sound sources.

From the plot it can be seen that Lights visualisation has the least amount of outliers and the

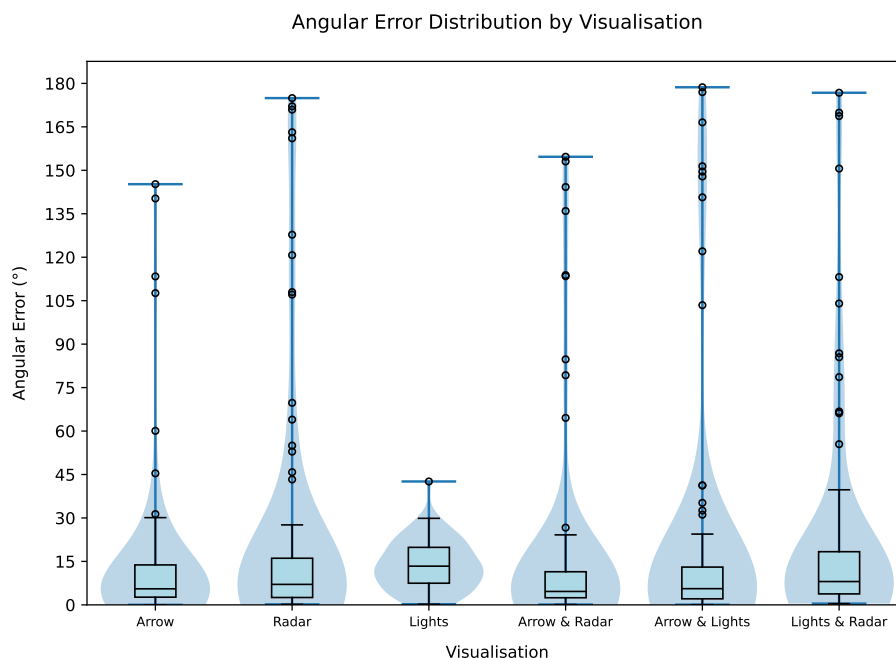


Figure 5.3: Violin box plot of angular error distribution by visualisation. It is clear that Lights has the least outliers but the highest median angular error. Arrow, Arrow and Radar, and Arrow have similar low median angular errors and small spread of values. Radar has a slightly higher median and larger spread but still smaller than Lights.

lowest maximum value. This suggests that the Lights effectively conveyed the area in which the sound source could be located and made it difficult for users to misunderstand the angle at which the sound source was located. This resulted in the Lights having the lowest mean of 13.7 degrees and standard deviation of 8.58 degrees, this is largely due to the standard deviations of the other visualisations being inflated by outliers, as seen from table 5.2.

However, from table 5.2 we can see that the Lights had the highest median of 13.36 degrees, suggesting that although the Lights conveyed the uncertainty range that the sound source could be in, it did not provide users with a means to accurately pin point the location of a sound source on the azimuth. Depending on how this system is being used, this trade-off may be preferable if precise interpretations of the sound sources angle are not necessary and users only wish to understand the broad area that the sound lies in to gain general awareness.

From 5.3 we can see that the three visualisations with the lowest medians were Arrow, Arrow and Lights, and Arrow and Radar. This suggests that the Arrow provides the most effective precise feedback in relation to the angle of the sound being located. This could be due to the fact that the Arrow is placed in the user's surrounding environment, and points to the actual direction in the users environment instead of needing to be interpreted and translated into a real direction as is the case with the head-mounted visualisations.

The combinations of visualisations also perform well in terms of angular error, with combinations including the Arrow visualisation having smaller interquartile ranges than the Lights and Radar combination suggesting that these combinations better represent the angle of detected sound to the user.

Angle against distance distribution

Table 5.2: Angular Error summary statistics by Visualisation

Visualisation	n	Mean(°)	Median(°)	SD(°)	Min(°)	Max(°)	Range(°)
Arrow	81	14.5	5.54	28.01	0.02	145.21	145.19
Radar	81	26.01	7.07	45.52	0.17	174.92	174.75
Lights	81	13.7	13.36	8.58	0.22	42.59	42.37
Arrow & Radar	81	18.39	4.63	36.75	0.11	154.67	154.56
Arrow & Lights	81	23.42	5.59	46.07	0.07	178.66	178.6
Lights & Radar	81	24.32	8.05	40.79	0.43	176.77	176.34

Table 5.3: NASA TLX summary statistics by visualisation.

Visualisation	n	Mean	Median	SD	Min	Max	Range
Arrow	9	3.46	2.5	2.69	0.83	8.0	7.17
Radar	9	3.43	2.17	2.92	0.5	7.67	7.17
Lights	9	4.04	3.67	2.41	0.5	8.83	8.33
Arrow and Radar	9	3.7	3.17	2.83	1.0	8.83	7.83
Arrow and Lights	9	3.5	3.17	2.22	0.5	7.5	7.0
Lights and Radar	9	3.72	3.0	2.48	0.67	8.83	8.17

Figure B.5 found in the appendix, shows a scatter plot of angular error against distance to target. It can be seen that as the distance to target decreases, the range of angular error values recorded increases drastically when the distance to the sound source is below 2.5 metres. This suggests that when users were within close proximity to sound sources, they were more likely to misjudge the angle of the sound source, resulting in large angular errors being measured.

It can also be seen that when using visualisations consisting of the Arrow or Radar visualisations, some participants misjudged the angle by around 180 degrees. A possible reason for this could be that these users were stepping past the sound source resulting in high angular errors near 180 degrees. This highlights the issue that the visualisations evaluated may need to better present near distance to sound sources to the user. This idea is further discussed in this chapter in results found from participant interviews.

This relationship between distance to target and angular error gives reason to the outliers present on Figure 5.3 for all visualisations, apart from Lights. This also suggests that at near distances to sound sources, Lights provided the most effective feedback to users.

5.4.2 Surveys

To further assess the effectiveness of each visualisation techniques ability to convey the location of spatial sounds, the results from the survey after each condition, and the final survey considering all conditions, were analysed.

NASA TLX Scores

To analyse the difference in perceived workload for each visualisation, Figure 5.4 shows a violin box plot of the NASA-TLX score distribution per visualisation. As a 21 point scale was used for each of the dimensions of the NASA-TLX, scores were calculated out of 21, with 0 representing extremely low workload and 21 representing extremely high workload.

Figure 5.4 shows that the interquartile range was largest for the Radar, suggesting that Radar was the most controversial in terms of workload, despite this the Radar had the lowest median of 3.43 as shown in Table 5.3.

It can also be seen from Figure 5.4 that the Lights had the highest median NASA-TLX Score, this suggests that this visualisation required the most cognitive effort to interpret, this could largely

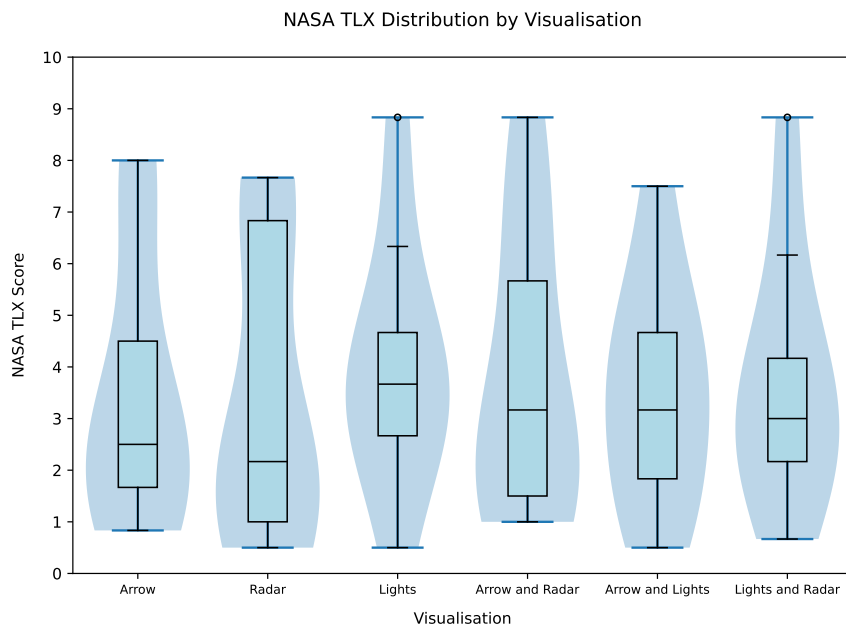


Figure 5.4: Violin box plot of NASA TLX Score Distribution by visualisation. All visualisations scores are in a range below 9. Radar has the highest range, largest spread of values, and lowest median. The other score distributions are similar with medium spreads and medians.

be due to the fact that this visualisation is the least explicit as it does not present the angle of the detected sound to the user only the general direction in which the sound is detected.

In addition, the large range of NASA-TLX scores for each visualisation also suggests that the perceived workload varied among participants, with ranges between 7.00 and 8.33 for each visualisation, as shown in 5.3. This indicates that the visualisations were not experienced uniformly in terms of workload, with some participants finding them relatively intuitive and low in demand, while others perceived a higher workload. This variation may be due to a number of factors such as, individual differences in interpretation, user preferences, or familiarity with Augmented Reality based visualisations.

Visual Load

From Figure B.8 in the appendix, we can view the responses to the questions surrounding visual workload and attentional impact for each visualisation.

It is evident that for the individual visualisations, respondents found the Arrow interfered with their awareness of their surrounding environment more so than the other individual visualisations. We can see that the individual Lights visualisation had the least amount of participants agreeing that the technique was visually demanding. From performance metrics we can see that it Lights obtained the highest medians for both distance to target and angular error whilst the Arrow obtained a low median for distance to target and angular error. This suggests that there may be a trade off between a visualisations visual demand and the effectiveness of it providing accurate feedback as to the location of sound sources. The Arrow is visually placed in the user's surrounding environment resulting in it interfering with the users environment, whilst the Lights is a head-mounted visualisation that uses subtle cues and interferes less, but in doing so may provide less accurate feedback as the location of the sound source.

It can also be seen that the pair combination of visualisations, Arrow and Radar, and Arrow

and Lights, have the largest amount of respondents agreeing that these visualisations were visually demanding, overwhelming, and interfered with their awareness of their surrounding environment. However, the Lights and Radar had similar respondents to the other individual visualisations, with the majority of respondents disagreeing that the visualisation was visually demanding, overwhelming, or interfered with their awareness. This is likely because Lights visualisation it self reported the lowest visual demand, with 89% of participants disagreeing that the visualisation was visually demanding. Due to the low visual demand perceived by participants the addition of Lights to the Radar visualisation did not interfere with the user's vision and visual demand to the same extent that the addition of other visualisations such as the Arrow did for example.

From these findings we can gather that visual demand is additive in nature, meaning combinations of visualisations that individually report low visual demand, result in visualisations that are comparable to individual visualisations in terms of visual demand.

Visualisation Rankings

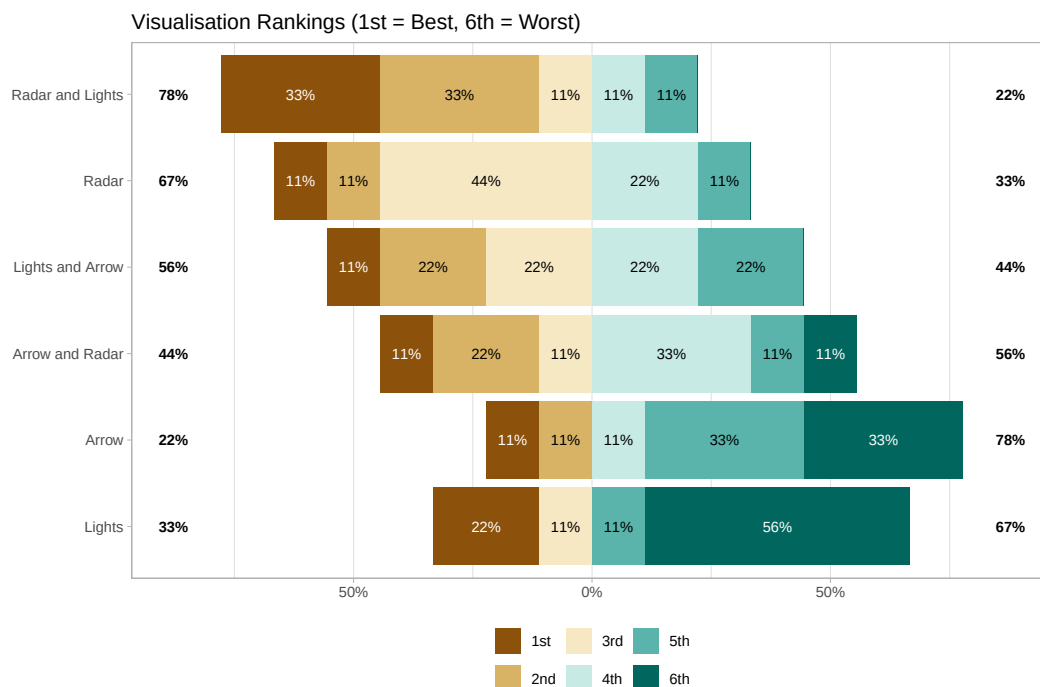


Figure 5.5: Plot of visual rankings showing the percentage of respondents that placed each visualisation in each ranking. Radar and Lights has the highest amount of 1st rankings with Lights having the highest amount of 6th rankings.

Figure 5.5 shows a graph of the distribution of participant rankings for each visualisation, ranging from 1st (the best) to 6th (the worst). From the graph it is clear that Lights received the lowest ratings over all with 56% ranking it as the worst visualisation. This suggests that the subtle cueing used in the Lights visualisation is not favoured by participants for localising sound sources. This could be due to the fact that the visualisation required more interpretation as it gave less explicit feedback regarding the angle of detected sound compared to the other visualisations.

Surprisingly, despite the Arrow's results in the performance metrics such as distance to target, and angular error, it was ranked as the worst visualisation by 33% of respondents, and the second worst by 33% of respondents. This suggests that although the Arrow enabled users to accurately

localise sound sources, many users did not prefer it, favouring other visualisations instead. The reasoning for this includes its high interference with the user's visual environment discussed in results above and in results from participant Interviews which will be mentioned shortly.

Radar and Lights had the highest proportion of favourable rankings from participants, with 33% of respondents ranking it as their best visualisation, and 33% of respondents ranking it as their second best. This suggests that this visualisation was generally well received by participants, and although the participant rankings do not directly indicate usability or performance, it was found from the performance metrics previously discussed that the Radar and Lights visualisation did prove effective. This indicates that this visualisation was appealing of the visualisations evaluated and that it may be the most appropriate to adopt for real world use. It is also evident that pair combinations of visualisations are favoured by many participants, confirming that multiple visualisations are a valid approach for visualising spatial sound.

It is important to note that for each of the visualisations at least one respondent ranked it as their most preferred visualisation. This highlights that there was no uniform consensus amongst participants as to how effective/user friendly these visualisations are, emphasising that user preference is varied, motivating flexible visualisations to be created.

Preference by distance

Figure B.3 in the appendix shows the distribution of preferred visualisations by participants at near, medium, and far distances. The combinations have been split and counted as individual visualisations due to some participants misunderstanding and responding with all visualisations they would find useful at different distances instead of what specific combination they would prefer.

We can see from Figure B.3 that 7 out of 9 participants chose Radar or some combination of visualisations involving the Radar, this demonstrates Radar's effectiveness at conveying near distance to users. Even so, some participants still preferred to use the Arrow or Lights, showing that user preference is mixed. As user preference is mixed, it is important to consider the flexible use between visualisations to accommodate for all possible user preferences.

At medium distances, preferences were evenly distributed, with each visualisation receiving five votes. This suggests that no single technique emerged as clearly preferable in this range, indicating a lack of consensus among participants regarding the most suitable visualisation for localising sound at medium range.

At far distances, Radar had the highest amount of votes, followed by Arrow and finally Lights. Although Radar had the highest votes, the difference in votes is small with only a difference of 2 votes between Radar and Lights, further suggesting that not only do users prefer different visualisations at different distances.

Radar was the most preferred visualisation overall amongst every distance apart from medium where there was an even distribution. This implies that the Radar is the most versatile visualisation and suitable to be used among all distances. This would incentivise the Radar to be chosen if only one individual visualisation could be implemented for use across a range of distances.

Figure B.4 in the appendix shows the visualisation choice per participant at each distance. The figure plots each participants choice at each distance and plots lines in between them. If all participants chose the same visualisations for each distance this graph would display flat horizontal lines. We can see from the graph that this is not the case as the lines are uneven and connected to different visualisations for each participant. This illustrates that individual participants were choosing different visualisations for each distance, confirming that the distribution found in Figure B.3 was the result of real distance dependent switching. It is important to still note that some participants did still choose the same visualisation for each distance, highlighting that for some, one visualisation was found to be most effective at all distances.

Confidence Per Visualisation

Figure B.7 in the Appendix presents the Likert scale responses to the survey question; "I felt confident using this visualisation to localise sound". From the figure we can see that the distribution of responses is similar across visualisations with Radar performing the best with all responses agreeing to the statement. The Arrow on the hand recorded the lowest proportion of agreement at 78% , suggesting that users felt the least confident when localising sound using the Arrow. This could be due to a number of reasons, one being that as the Arrow is the largest visualisation, the movement of the arrow affected by the uncertainty of the simulated microphone array is the most visually perceptible, causing users to interpret the Arrow as proving less precise and reliable measurements, even when its localisation performance in terms of performance metrics was objectively strong.

Scenarios

Participants were asked questions considering 4 different scenarios representing different contexts where hearing is impaired along with varying safety risks involved. For each of these questions, participants considered the system being used with a pair of modern smart glasses capable of full Augmented Reality. Scenario 1 was : Working in a safety critical warehouse with hearing protection on. Scenario 2 was : Walking along a busy road with headphones on. Scenario 3 was : Responding to a fire alarm in a large building. Scenario 4 was : Sitting in a cafe with headphones on.

Two questions were asked for each scenario with participants responding with a 7 point Likert scale from strongly disagree to strongly agree. Question 1 being ;The system would be useful in this Scenario, and Question 2 being ; I would choose to use the system in this Scenario.

Figure B.6 found in the appendix, shows the proportion of Likert responses for each question regarding the scenarios mentioned above. It is evident that participants considered scenario 1 to be the most suitable for use of the system, with 89% agreeing that the system would be useful, and 67% of participants agreeing that they would personally use the system in that context. This suggests that for high risk environments where maintaining awareness is important, participants recognised the potential benefit of using the system to inform them of the surrounding aural environment.

For scenario 2, 50% of participants agreed and 50% disagreed to each question with the level of agreement and disagreement varying. As this scenario involves walking along a busy road, this suggests that some users would not find this system beneficial and possibly distracting for busy public environments where there is some safety risk involved but not to the same extent as the first scenario considering a safety critical workplace. Participants also mentioned in the following interviews that they feared that the system would be overwhelmed by the amount of different sound sources present on a roadside and not be able to properly sense the direction of important sounds, instead distracting them and reducing their awareness.

For scenario 3 44% of the participants strongly disagreed that this system would be useful and that they themselves would choose to use the system in the scenario. Some participants discussed their reasoning out loud as they completed the survey and explained that this was because the system would point you towards the fire alarm itself, which in this scenario they would not find useful as they require directions out of the building not towards the fire alarm. This relates to the theme; Trust of system in noisy environments, discussed later in this chapter in results from participant interviews. There may be scenarios such as this one where safety is a concern but the location of the loudest sound in the users environment is not important to them. It also highlights the issue of the visualisation of sound source localisation being mistaken for explicit navigation directions in scenarios such as this one, possibly motivating the design of these visualisations to not directly replicate common navigation visualisations such as the Arrow.

For scenario 4, 56% of respondents disagreed that they would find the system useful and that

they would choose to use the system. For the first question, 33% strongly disagreed, suggesting that for some people, using the system in a relaxed indoor environment such as a cafe would provide them with no useful information about their aural environment, or that they simply would not require this information. In contrast, some participants did agree that the system would be useful, also agreeing that they would use it themselves, this indicates that this system may prove beneficial when being used in contexts where there is no major safety risk involved, and instead can help users maintain general awareness of their surroundings. As this scenario involves users sitting in a coffee shop, it is also possible that users may find the system useful to regain social awareness that has been lost due to a hearing impairment, for example if a person is trying to speak to you but is out of sight, this is discussed in the next section when discussing possible use cases discussed in participant interviews.

5.4.3 Interviews

To gather and analyse patterns from the semi-structured participant interviews conducted, thematic analysis was performed on the interview transcripts using Braun and Clarke's thematic analysis framework (Braun and Clarke 2006). The transcripts were read over and reviewed for familiarisation, then using deductive and inductive codes all transcripts were coded. The codes were then grouped and refactored, with the transcripts being recoded with the final refactored codes. From the groupings of codes, seven themes emerged; distraction from Arrow, passivity and ease of use of Lights, encoding distance, need for flexible use, need for classification to filter sound, trust of system in noisy environments, and possible use cases.

Distraction from Arrow

Many users found the Arrow distracting, suggesting why it was ranked the worst visualisation by 56% of participants, as shown in Figure 5.5.

In the interview, Participant 6 said "Um, because when you get too close to the objective with Arrow, it spins away from you, and it's much harder to see when it spins away from you, because you have to, like, pivot all the way to see where it's gone. Whereas the Radar, you can just still see it in your same, like, in front of you, and it'll turn around, just tell you that you've gone too far, but you can see it right from you.". Participant 8 also stated "Yeah, I didn't like it when you each time you restart, you've got to spin around and find it again."

As the Arrow is placed in the user's surrounding environment relative to the position of the user's body, if the sound source is outside of the user's field of view, so too will the Arrow. This makes it difficult for users such as participant 6 and participant 8 to quickly respond and interpret the direction of sound sources that are outside of their field of view. This can also occur repeatedly when users get within near distance of sound sources and pass them, suggesting why angular error increased whilst using the Arrow visualisation as users got closer to the sound sources, as shown in Figure B.5.

Participants also found the Arrow to occlude their surroundings and distract them from their surrounding environment. For example, when asked why participant 1 chose Arrow as the worst visualisation, they stated "I'm not quite sure with specifics, but I think the Arrow as I said might sometimes be a bit distracting if it was. an environment where you had to be very careful where you were."

Similarly, When asked why participant 2 chose the Arrow and Radar as the worst visualisation, they stated "The Arrow, although useful, and, like, very clear as to what was going on, was just too visually, like, stimulate. Just massive." This quote highlights that the effective localisation capabilities of the Arrow, found in the results of the performance metrics, do not correlate with user preference.

Participant 5 also mentioned that they found the Arrow's representation of the direction of sound to be too precise, putting forward the idea that the Arrow may be too explicit in its presentation

of the location of the sound source. They stated "The Arrow was pretty big and I think I took context into consideration your surroundings quite a lot. When you see the Arrow, I feel like it's very easy to just directly just follow straight away which way you're going with the Arrow, but with the Radar is kind of easier to tell which general direction, if you get what I mean." They went on to say "But the Arrow was maybe taking too much cognitive load off of you. So you're actually kind of not really thinking about it any more. You just see an Arrow automatically, you think, I'm just going to follow it this way, right? And then when you go to a barrier, because the Arrow's not showing you navigation directions its showing you the direction. Whereas I would say the Radar is more obvious that this is the general direction you want to go in. " Participant 5 makes it clear that the Arrow's direction can be confused for actual navigation instructions, leading to confusion when it guides one to a barrier. This suggests that the Arrows affordance is one of navigation directions and not of general location of sound. This highlights an important possible downside to deriving visualisations from common AR navigation visualisation techniques, they can be confused for actual navigation directions.

However, some participants found the Arrows explicit direction in the real world to be helpful with participant 4 stating "So the Arrow provided more like real world input of where it was and also the Lights made so if I turn my head, I could kind of get like. A sense of where it could be like just like I was if my head was like a Radar."

Some participants such as Participant 2 thought that not only the Arrow but all visualisations would be distracting in high focus scenarios "I think if you're doing something with like high concentration. So you know if you were. Not high concentration in the sense of studying or writing something, but concentration in the sense of obviously this is very specific like a surgery."

However, some thought the visualisations would be too distracting all together with participant 5 stating when asked if they would use these visualisations "Yeah, I think, um, That would probably be too busy and just too, like, chaotic for my eyesight". This suggests that for some having any occlusion of vision is too much or visualisations should be designed that are more passive and less occluding.

This theme found from analysis of the interview suggests that although the Arrow may perform well in terms of performance metrics such as distance to target and angular error, it is likely to be distracting in real world use, can be confused for actual navigation directions, and hinders users ability to locate sources out with one's field of view. This is likely to be a reason why so many participants ranked it as their worst visualisation. This suggests that despite the fact that a visualisation technique may result in accurate localisation, it is not necessarily found usable by users.

However, like with all visualisations evaluated in this project, some participants did prefer the Arrows explicit direction, highlighting the need to provide users with a choice of visualisations for flexible use.

Passivity and ease of use of Lights

Many participants found the passive nature of the Lights to be useful, likely giving a reason why Radar and Lights were ranked so highly by participants.

Participants discussed how they found the Lights to be the least visually demanding with Participant 9 discussing how they would use this visualisation to gain added awareness in everyday life. They said when asked if they would use the system in everyday life given it was implemented into a pair of smart glasses "Yeah, I would. I think if I was just going about my daily life and I wanted something that could notify me of something, but not necessarily that I'm looking for, just a general sort of extra awareness, ... I think that the Lights feel more passive." This quote highlights the benefit of the Lights passivity, it makes the visualisation more appealing for adoption in everyday life due its low visual demand.

When asked why Participant 1 chose Radar and Lights as the best visualisation, they said "It felt and the Lights the like the Radar Lights felt me feel kind of stable. They made me feel like I could very quickly realise what direction I should be facing and by having something on both sides instead of just in the centre.". This user shows that the combination of Lights with the Radar visualisation provided substantial visual feedback that made them feel confident in localising sound quickly.

As the noise from the uncertainty of the microphone array was not always visible with the Lights visualisation some participants felt more confident using it. When asked why they would use Lights at medium and far distance Participant 2 said "It was like, I already kind of knew what direction to go in from far away. But with with the other two, it was like, just not sure, yeah, just not sure it was still moving about a lot. There's but the Lights it was really clear I need to be facing this way. "

Participant 5 found the Lights useful at close range stating "Uh, and then I think the Lights were pretty useful for like when you actually got right to the sound because. Well, the other one's they they spun out like a compass, but yeah, I guess you kind of knew which way exactly to look because one side would one side would go off. But when you're actually looking for confirmation of like this is definitely the sound, then the Lights were useful, I think. Because you know what the final state, you know, the exact final state that you're looking for is like the exact, the complete Lights." This suggests why there are no outliers for Lights compared to other visualisations in Figure 5.3.

Some participants come to the consensus that Lights was suited for far and medium distances however with Participant 3 stating "So the Lights were my favourite, but when you're far away, obviously, you, when you get that big cone from the Lights, you're maybe looking around for a bit to kind of find where the Radar is super clear where it takes that away."

This proved to be a common downside of the Lights with 6 out of the 9 participants reporting in interviews that it did not give them confidence in finding the specific point the sound was located. Participant 9 stated "I think as sort of natural as it looked in terms of it was the least visually distracting, I think it was also my least confident in terms of being able to actually pinpoint the sound to a specific point."

Encoding distance

Many participants mentioned in the interviews how they were unsure of when they were near to sound sources, with many suggesting encoding arms-reach distance to detected sound sources in order to aid localisation.

Participant 9 said "Sort of rely on the system and then maybe like a colour change when it reaches sort of 100% or like some sort of verification that it's hit a ceiling. Yeah. And then I think with the Arrow, I would have thought maybe a more, again, kind of similar, either a more vibrant colour change or like colour intensity or a colour change altogether". This quote emphasises the participants understanding that a near distance would result in a colour change, more drastic and noticeable than the current increase in opacity.

Participants did not only suggest that this near distance be encoded using colour. For example, Participant 1 suggested an audio cue, stating "For like the nearby one, so when you get really close, I think it would be great to have an audio cue, even though I know it might overlap with that sound from the surroundings anyway." They highlight an issue with using audio cues as they may not be heard if one's hearing is impaired from their surroundings. This modality may prove effective if the users hearing is impaired by a pair of noise cancelling headphones where sound can be played through them whilst keeping the surrounding noise impaired.

Participant 1 also suggested having the visual flash to convey close distance, "Yeah, just to kind of, like I was saying with a car, so it kind of you see both the Lights and the beeping at the same time. And it kind of gets across the sense of urgency is how close you are to something. I

think me personally that would make it more effective for me." This analogy of a car reversing highlight an important aspect of design, affordances. Making sure that representations of distance correspond with what users are familiar with in their every day life is important.

A number of participants found the Radar to be the best for conveying when they were very close to a sound source, as shown in Figure B.3 in appendix B. Several participants used the Radars fast moving notch when close to a sound in order to discern near distance. When asked why they chose Radar for near distance, Participant 2 said "Yeah, well, because it went so crazy when you were kind of at the right spot, you just take that one step back and then it's like. So I just yeah, I thought the Radar was quite good for that."

As the Radar is implemented as a head up display, it also informs users of the direction of sound arriving at all angles, this enabled participants to quickly locate sounds sources. Participant 4 stated in their interview "And then when you're really close to the target, I said Radar because Radar when you go in front of the target so the target's behind you. The Radar shows you that it's behind you. Like perfectly. So it's to really pinpoint and get closest to the target. Radar was the best when you're really close."

Need for flexible use

From asking why participants used certain visualisations at near, medium and far ranges of distance, many participants suggested that the system should allow users to swap between visualisations depending on the distance from the sound source. For example, Participant 4 stated in their interview "For visualisation, I think the closer you get to the target, maybe swap modes like I said, like the Radar." Participant 4 also said "So Arrow and Lights at far ranges, Arrow really pinpoint where exactly the target is at long ranges."

Participants also highlighted the need for flexible use and swapping between visualisations in terms of usage context. Participant 10 said "As nonintrusive as possible and also completely invisible if I decide it to be, for example, like, yeah, modular, I guess, or sort of, you can increase the, you know, and maybe I don't want anything on my, in my periphery or in my screen and maybe I want to have that extra awareness because I'm listening to music and then maybe I'm doing a specific task, so therefore I would prefer having a Radar or an Arrow, but also the option to have none or some or partial, you know, very customizable." This emphasises that not only do users find the visualisations differ in terms of effectiveness at different distance ranges but also suitability across different usage scenarios with Participant 10 above mentioning they would find the Lights useful in relaxed scenarios for extra awareness, and they would use the Arrow or Radar in more engaged tasks possibly where one may be attempting to locate sound sources.

Need for classification to filter sound

Several participants stated that the labelling/classification of detected sounds would prove beneficial.

Participant 5 mentioned "I guess if the system could be used for like a focus sound, that would be pretty useful in like a noisy environment like. If it was able to pick up, I know right now just uses a like any sound, but if it was able to tune into like a focus sound." This participant describes the system being able to only display a certain type of sound, through classification of sound this could be made possible.

When asked "Are there any scenarios you can think of where this would be of use to you?", Participant 7 responded "Maybe with changes to label sounds, it could be useful sometimes". Participant 7 describes here that labels paired with the detected sounds would be useful, this suggests that by showing the classification labels of detected sound sources, users could gain a better understanding of their surrounding aural environment, better informing them of sounds they may need or want to pay attention to.

This use of classification has been achieved in other wearable sound source localisation solutions

by running classification models on the audio streams to filter the feedback given (Daoud et al. 2015). Labelling of detected sounds has also been explored but presentation of labels has been chosen empirically and usability is still to be evaluated (Shen et al. 2012; Guo et al. 2020).

As many participants required some sort of classification to improve the usability of the system, further work on how these visualisations can be adapted to show classification labels could be done.

Trust of system in noisy environments

Participants often noted their concern about the performance of the system in noisy environments where many sounds compete in loudness.

When asked who they think would benefit from the system, Participant 7 said "I guess it depends how reliable it is at picking up everything in bit loud environments because maybe if there was too much noise it couldn't pick up something.". This excerpt displays how adoption of this system may rely on its ability to perform in loud environments, benchmarking and testing would be required in order to assess its ability at detecting sounds in this environment and possible improvements may be made to filter the incoming audio into the system.

When suggesting improvement for the system Participant 4 mentioned this possible filtering of sound noting, "Have it where I can like, Filter noise out and like so. So if there's like an ambient sound going on, let's say like a a washing machine or like a fan going on. It will know how to filter it out". Filtering could be a method at improving the systems performance in environments with many loud sounds or simply improve its usability by not visualising prominent background sounds such as a washing machine as the participant mentioned above.

Adding onto this idea, when discussing possible use of the system Participant 3 mentioned "I think the only one I maybe disagreed with was in the cafe, just 'cause there's a lot of audio information and not much important audio information. It would just be like, Idle chit chat. Like, you know, background noise and that could maybe just kind of make it go a bit haywire, but I guess you would just turn it off like that.". Here they add onto the lack of trust of the system in noisy environments, worrying that the systems may visualise a mess, possibly in the form of many garbled angles and distances. They also touch on the point that they would only want to have sounds important to them be visualised, background noise is not something they believe would benefit them.

Possible Use Cases

During the interviews possible use cases of the system were discussed. The use of the system to benefit those who experience overstimulation was discussed with Participant 3 suggesting "And people, I guess, as well, neurodivergent people who have some kind of, auditory overstimulation. Like, that they would maybe very often be wearing hearing protection". Participant 1 further explored this testifying to their own experience "Or. Maybe sometimes I get quite overstimulated with my environment and I like more. Simple surroundings that it felt quite calming so I feel like it would help me".

Some participants stated that they would use the system to gain awareness about their surroundings. Participant 4 discussed this saying "In a coffee shop, if I'm listening to music and I've got like noise cancelling on and someone's trying to talk to me, I think like a nice pop-up or like visualisation that someone's talking to would be good.". This displays how the system can retrieve social cues that may have been lost due to impaired hearing, emphasising the potential of the system to improve social awareness.

Participants also mentioned use of the system in safety critical scenarios. Participant 3 said "I think maybe in just high danger environment. You would need like really quick Response and feedback to kind of like dangerous, like sounds that indicate danger.". This indicates that the system may alert users of possible warning sounds but highlights that the systems response would have to

be very quick if being used for such a reason. Benchmarking of the latency involved in a final prototype would need to be measured, at the moment measurements of sound are networked from the laptop to the headset resulting in latency that could be greatly reduced if a data was transferred via a cable instead or computation was done locally on the headset or wearable display this system was implemented into.

Regarding safety critical where one's hearing is impaired Participant 8 said "I think there could also be that. It's obviously like safety aspects to it so. I don't know maybe firemen in a building on fire. Yeah. To be aware of sounds that they can't hear themselves. ... Its that sort of thing where you can't really do it with your own vision, because there's stuff blocking it.". This highlights the systems ability to aid sound localisation where visual aids are not present.

Participant 8 said "On a bike, it might be useful. If you could know that there's a car or a lorry coming up behind you, and how close it's getting, and that sort of thing.".

More relaxed use cases were also discussed with Participant 8 noting "Yeah, I think if you were like bird watching or out, looking for animals, and it could say there's a fox behind you or something. I think that would be really useful. You can't see it, but the system can hear it." The participant describes the system being able to classify sounds to further inform them of the sounds in their environment.

5.5 Analysis and Discussion

By assessing the results from performance metrics, surveys, and interviews, this section addresses the two research questions defined in this chapter, examining which visualisation techniques proved most effective at representing spatial sound, and how individual visualisations compare to pair combinations.

RQ1: Which visualisations are most effective for conveying spatial sound

The results demonstrated differences between the three individual visualisation techniques in terms of multiple attributes of effectiveness, with no single visualisation technique emerging as the most effective for all participants. User preference of visualisations was varied with each visualisation being most preferred by at least one participant.

The Arrow showed the strongest objective results in terms of performance metrics, achieving the lowest median distance with the lowest standard deviation. The Arrow also achieved the second lowest median angular error and the second lowest standard deviation. These results likely stem from the fact that as this visualisation is placed in the users surrounding environment, it points directly to the detected sound source requiring no translation by the user to interpret the actual direction towards the sound source resulting in accurate localisation of sound sources.

Despite the Arrow's strong results in performance metrics, the Arrow was ranked as the worst or second worst visualisation by 66% of participants, indicating that there is a significant disconnect between localisation accuracy and perceived usability of visualisation techniques. It was found from participant interviews that many participants found the Arrow to be distracting, with many noting that it occluded their surrounding environment. It was also found that when using AR navigation visualisations such as the Arrow, the direction displayed presented may be confused for actual navigation directions as this is a common known affordance of common visualisation techniques used in AR navigation. This leads to confusion when barriers in the environment are met, indicating that the use of common AR navigation visualisations, such as the Arrow (LYNAM et al. 2024), can cause confusion amongst users and is possibly inappropriate for presenting the location of spatial sounds. Furthermore, it was found that the Arrow reported the lowest proportion of participants agreeing that they felt comfortable localising sound using the visualisation, with 78%, compared with the Radar which had an agreement of 100%. This

could possibly be due to the fact that the Arrow was the largest visualisation evaluated, meaning that the simulated microphone array's localisation uncertainty was most perceptible in the Arrow causing participants to interpret it as providing the least precise and accurate measurements of the sound source location.

It was also found from participant interviews that users did not prefer the Arrow as it does not preserve the omnidirectional nature of hearing and disappears from view when a sound source is detected outside of the user's field of vision. This caused users to become frustrated and likely contributed to the increases in angular error observed at close distances, as shown in B.5. It is evident that it is important to design visualisations of spatial sound that preserve the omnidirectional nature of hearing, sounds are heard from all directions, so all detected sounds should be displayed to the user, as the Arrow visualisation did not achieve this many users found it difficult and frustrating to use when localising sound as found in participant interviews. The other two individual visualisation techniques created, Radar and Lights, maintained the omnidirectional nature of hearing.

The Radar visualisation achieved a median distance to target of 0.80m with a standard deviation of 1.85m, these results being similar to the median and standard deviation of distance to target reported with the Arrow. Its median angular error was slightly higher than the Arrow's at 7.07 degrees with a standard deviation of 45.52. Whilst the performance metrics reported for the Radar were slightly higher, it was preferred by more participants, resulting in it receiving the highest proportion of positive user rankings amongst the individual visualisation, and rated the most highly in terms of confidence at a 100% agreement that users felt confident localising sound using the visualisation. As this visualisation is presented relative to the users head position, the direction at which sound is detected is always visible to the user. This addresses a core limitation of the Arrow. From interviews, participants also found the movement of the Radars notch useful for conveying the near distance of sound sources.

The Lights visualisation achieved the lowest mean angular error of 13.7 degrees with the lowest standard deviation of 8.58 degrees, this was due to there being no extreme outliers in angular error, suggesting it effectively prevented any misunderstanding as to the sound sources direction. Despite this, Lights had the highest median angular error of 13.36 degrees as well as the highest median distance to target of 1.86m. This indicates that whilst it prevented the direction of the sound source from being misunderstood, it did not afford precise localisation. This correlates with the design of the visualisation as it encodes only three directional states rather than a continuous angle, meaning the broad region in which a sound source lies is presented instead of its specific direction. Participants reflect this in interviews, with many noting that they felt insufficiently confident using the light to pinpoint the location of a sound source especially at farther distances. The Lights also reported the highest proportion of respondents disagreeing that the visualisation was visually demanding (89%). The passive nature and low visual demand of the Lights appealed to many participants, with them stating in interviews that they would be most likely to adopt this visualisation to use in order to gain awareness in their everyday life.

The Lights reported low visual demand paired with less precise localisation capabilities, whilst the Arrow reported higher visual demand paired with precise localisation. This suggests a perceptual trade-off with more explicit, precise visualisations requiring greater visual load, while subtler ambient directional cues require less but at the cost of localisation accuracy. This indicates that the choice of visualisation used depends on the usage context and the user needs: whether they require precise localisation of sounds, or simply a general awareness of sound direction with minimal visual distraction.

Quinn and Gabbard (2024) proposed a design guideline for visualisations of out of view objects, this design guideline stated that visualisations should use subtle cues instead of explicit ones where possible and that this would not likely affect search performance. From our results it can be seen that the subtle cues used in the Lights visualisation resulted in poorer localisation performance,

whilst the more explicit Radar visualisation reported better localisation performance, suggesting that this design guideline does not hold true for visualising spatial sound.

Overall, as Radar was found to be the most versatile individual visualisation across both performance and usability, our findings suggest that out of view object visualisations that are head-mounted and provide explicit cues as to the location of sound, such as the Radar, are the most versatile and suitable for use when localising spatial sound. Lights on the other hand may be most suitable for passive awareness as it reported the lowest visual demand with participants discussing in interviews that this visualisation would be the most suitable for their everyday use cases. An important risk was highlighted that common AR navigation visualisation techniques have strong existing affordances for navigation and can cause confusion when used in a sound localisation context, suggesting that common AR navigation visualisations are not suitable for conveying the location of spatial sound.

RQ2: Are combinations of visualisations effective for conveying spatial sound

The pair combinations of visualisations performed similarly to the individual visualisations in terms of performance metrics, with higher reported visual demand and interference with awareness.

Whilst a higher proportion of participants reported the information presented by the combination of visualisations to be overwhelming, LYNAM et al. (2024)'s concern that combinations may cause information overload and be unusable did not occur, with combinations performing well in terms of performance metrics and perceived workloads. From participant rankings of visualisations, Radar and Lights emerged as the most preferred visualisation overall with 33% of participants ranking it first, the highest proportion for any visualisation. This combination of Radar and Lights also reported the lowest visual demand and interference with awareness of the combinations evaluated. As Lights reported the lowest visual demand of all the visualisations, this suggests that visual demand may be additive in nature suggesting that combinations should comprise of at least one visualisation that requires little visual demand such as the Lights. This also suggests that Quinn and Gabbard (2024)'s recommendation of creating combinations of visualisations that do not create visual clutter holds true for enabling effective localisation of sound.

Unsuitable Contexts of Use

From analysis of the interviews, the theme of distrust of use in noisy environments emerged, this highlights participants concern that the system may struggle to reliably interpret and visualise complex or cluttered auditory environments, potentially providing inaccurate and misleading representations of the surrounding aural environment, posing a safety concern. These findings suggest that systems which sense and visualise locations of spatial sounds clarify the limitations of sensing capabilities, such as possible noise thresholds or reduced performance in complex auditory environments, so that users can better understand when to trust the systems feedback provided.

5.5.1 Design Guidelines

Based on the findings discussed above and from themes found from analysis participant interviews, the following design guidelines are proposed for creating effective visualisations of spatial sound in augmented reality.

Preserve the omnidirectional nature of sound

Sound is heard from all directions, visualisations that do not convey the direction of sound outside of the users fail to inform the user of their environment the same way that hearing might do. It became clear from participant interviews that many participants found the Arrow frustrating

to use as it did not adhere to this principle. Head-mounted visualisations that remain visible regardless of the users head position and field of view are best suited for representing spatial sound detected from all directions.

Combine complementary visualisations Combinations of visualisations that are similar in terms of explicitness and high in visual demand become redundant when combined, instead we recommend that explicit visualisations such as the Radar are paired with passive visualisations such as the Lights that provide subtle cues with low visual demand.

Support flexible and adaptable visualisations It was found that individual participants preferred different visualisations across varying distances as well as use cases. As user preference of visualisations was also varied overall, it is important to provide users with a range of flexible visualisations that can be adapted by the user to suit their needs across multiple usage contexts.

Avoid directly using visualisations commonly used for navigation

It was found that common navigation visualisations such as the Arrow, may be falsely interpreted as explicit navigation directions due to the known affordances of such visualisations. This risks users becoming frustrated and confused when following the explicit direction of the Arrow and reaching barriers and obstacles instead of the sound source. Visualisations that more clearly communicate the detection of sound, such as an overview approach found in the Radar, may be more appropriate.

Present near distance explicitly

Participant interviews highlighted that they found it difficult to understand when they were in close proximity to sound sources, with many suggesting an explicit clear encoding of near distance as a solution, such as a colour change. Visualisations should be designed with a clear change of state visible to the user when near distance is measured. By addressing this issue in the design of visualisations, users could better interpret near distance likely reducing the increase in angular error observed at close ranges.

6 | Conclusion

6.1 Summary

Sound is an important aspect of one's environment, hearing enables us to detect, classify, and localise sounds in our surroundings. When hearing is impaired, this may be situationally through the use of hearing protection or noise cancelling headphones for example, the ability to localise sound is worsened resulting in a loss of awareness of one's surroundings.

To minimise the impact of impaired hearing, we developed a prototype that visualised the location of spatial sounds in one's aural environment using augmented reality. Three visualisations were created, derived from visualisations found in augmented reality solutions to navigation as well as out of view object visualisations. The location of sound was detected by processing the audio signals of a miniDSP UMA-8 microphone array. A microphone array is a set of microphones placed in a specific geometry, the audio signals from these microphones can be used to compute the direction of arriving sounds, these methods are called Direction Of Arrival (DOA) methods. An algorithm was developed that used the SPR-PHAT algorithm to detect the direction of sound sources, it also used the root mean square of the microphone arrays audio signals to compute the approximate distance of arriving sounds.

The microphone array was mounted upon the Meta Quest 3 virtual reality headset, and the approximate distance and angle of sound was networked to the headset and visualised using the 3D game engine Unity.

We used the prototype to conduct an evaluation which aimed to answer two research questions asking; what visualisations are most effective for conveying spatial sound (RQ1) and are individual or pair combinations of visualisations more effective for conveying spatial sound? (RQ2).

The evaluation was conducted using a simulated virtual environment where the sensing capabilities of the microphone array were replicated. Participants were tasked at localising sounds in a 3D warehouse environment. The evaluation had six conditions, three being the individual visualisation techniques and three being the pair combinations of the three individual techniques. Performance metrics such as the distance to the sound source and angular error were measured to evaluate the effectiveness of each visualisation. Participants completed surveys to measure the perceived workload, visual workload and attentional demand, and confidence when using each visualisation. User preference and context of use was recorded through surveys as well. Finally a semi-structured interview was conducted with participants to gain insight into their reasoning into how each visualisation impacted their localisation of sound sources.

In regards to RQ1 It was found that user preference amongst visualisations is varied, however the combination of Radar and Lights emerged with the highest proportion of users preferring this visualisation. Regarding RQ2 it was found that combinations of visualisations performed similarly to individual visualisations and were preferred by many users meaning that combinations of visualisations prove effective for visualising spatial sound. Finally design guidelines derived from our findings were proposed, enabling the design of effective visualisations of spatial sound.

6.2 Future work

There are many areas to further explore stemming from the implementation and evaluation of the prototype developed in this project.

Firstly, as a simulated environment was used in this project, the real capabilities of the microphone array and real world navigation were likely not fully replicated meaning the generalisability of our results to real world use is limited. Using the prototype in a real world environment to evaluate the use of visualisation techniques would garner more reliable results as the actual performance of the prototypes sound localisation capabilities would impact how effective each of the visualisations are.

Additionally, one's aural environment is commonly made of multiple sound sources whilst this project only visualised one source at a time. By adapting these visualisations to convey the location of multiple sound sources at once, a better understanding of how effective these visualisations are at representing a richer, more realistic aural environment can be understood.

The visualisations created also only visualised the azimuth of sound sources as the DOA algorithm implemented in this project could only detect sound on the azimuth. By visualising the elevation of sound sources as well, a better representation of an individuals real aural environment could be conveyed to the user, with more accurate representations of the location of sound sources possibly benefiting localisation. As humans also less capable at detecting a sounds elevation compared to its azimuth (Ege et al. 2018), this could provide benefit beyond the capabilities of sound source localisation relying only on human hearing.

Another area that this project did not cover is sensing moving sound sources. This project only evaluated sensing stationary spatial sounds, in real world environments it is common for sound sources to be moving, for example cars and people are often moving whilst emitting sound. Evaluating the visualisations ability to convey moving sounds would give a better understanding of how suitable these visualisations are for real world environments and if adaptations must be made to accommodate for this change.

A | Ethics Checklist

**School of Computing Science
University of Glasgow**

Ethics checklist form for 3rd/4th/5th year, and taught MSc projects

This form is only applicable for projects that use other people ('participants') for the collection of information, typically in getting comments about a system or a system design, getting information about how a system could be used, or evaluating a working system.

If no other people have been involved in the collection of information, then you do not need to complete this form.

If your evaluation does not comply with any one or more of the points below, please contact the Chair of the School of Computing Science Ethics Committee (matthew.chalmers@glasgow.ac.uk) for advice.

If your evaluation does comply with all the points below, please sign this form and submit it with your project.

-
1. Participants were not exposed to any risks greater than those encountered in their normal working life.

Investigators have a responsibility to protect participants from physical and mental harm during the investigation. The risk of harm must be no greater than in ordinary life. Areas of potential risk that require ethical approval include, but are not limited to, investigations that occur outside usual laboratory areas, or that require participant mobility (e.g. walking, running, use of public transport), unusual or repetitive activity or movement, that use sensory deprivation (e.g. ear plugs or blindfolds), bright or flashing lights, loud or disorienting noises, smell, taste, vibration, or force feedback

2. The experimental materials were paper-based, or comprised software running on standard hardware.
Participants should not be exposed to any risks associated with the use of non-standard equipment: anything other than pen-and-paper, standard PCs, laptops, iPads, mobile phones and common hand-held devices is considered non-standard.

3. All participants explicitly stated that they agreed to take part, and that their data could be used in the project.

If the results of the evaluation are likely to be used beyond the term of the project (for example, the software is to be deployed, or the data is to be published), then signed consent is necessary. A separate consent form should be signed by each participant.

Otherwise, verbal consent is sufficient, and should be explicitly requested in the introductory script.

4. No incentives were offered to the participants.

The payment of participants must not be used to induce them to risk harm beyond that which they risk without payment in their normal lifestyle.

5. No information about the evaluation or materials was intentionally withheld from the participants.
Withholding information or misleading participants is unacceptable if participants are likely to object or show unease when debriefed.
6. No participant was under the age of 16.
Parental consent is required for participants under the age of 16.
7. No participant has an impairment that may limit their understanding or communication.
Additional consent is required for participants with impairments.
8. Neither I nor my supervisor is in a position of authority or influence over any of the participants.
A position of authority or influence over any participant must not be allowed to pressurise participants to take part in, or remain in, any experiment.
9. All participants were informed that they could withdraw at any time.
All participants have the right to withdraw at any time during the investigation. They should be told this in the introductory script.
10. All participants have been informed of my contact details.
All participants must be able to contact the investigator after the investigation. They should be given the details of both student and module co-ordinator or supervisor as part of the debriefing.
11. The evaluation was discussed with all the participants at the end of the session, and all participants had the opportunity to ask questions.
The student must provide the participants with sufficient information in the debriefing to enable them to understand the nature of the investigation. In cases where remote participants may withdraw from the experiment early and it is not possible to debrief them, the fact that doing so will result in their not being debriefed should be mentioned in the introductory text.
12. All the data collected from the participants is stored in an anonymous form.
All participant data (hard-copy and soft-copy) should be stored securely, and in anonymous form.

Project title Visualising Spatial Sound in Augmented Reality: Evaluating Visualisation Techniques for Sound Source Localisation

Student's Name Joe Weir

Student Number 2770814w

Student's Signature Joe Weir

Supervisor's Signature Mark McGill

Date 16/03/2026

B | Figures



Figure B.1: A photo of the benchmarking taking place, with a microphone array placed one metre away from a Bluetooth speaker. The microphone array sits upon a compass which was used to measure the interval of angles to place the speaker at relative to the microphone array.

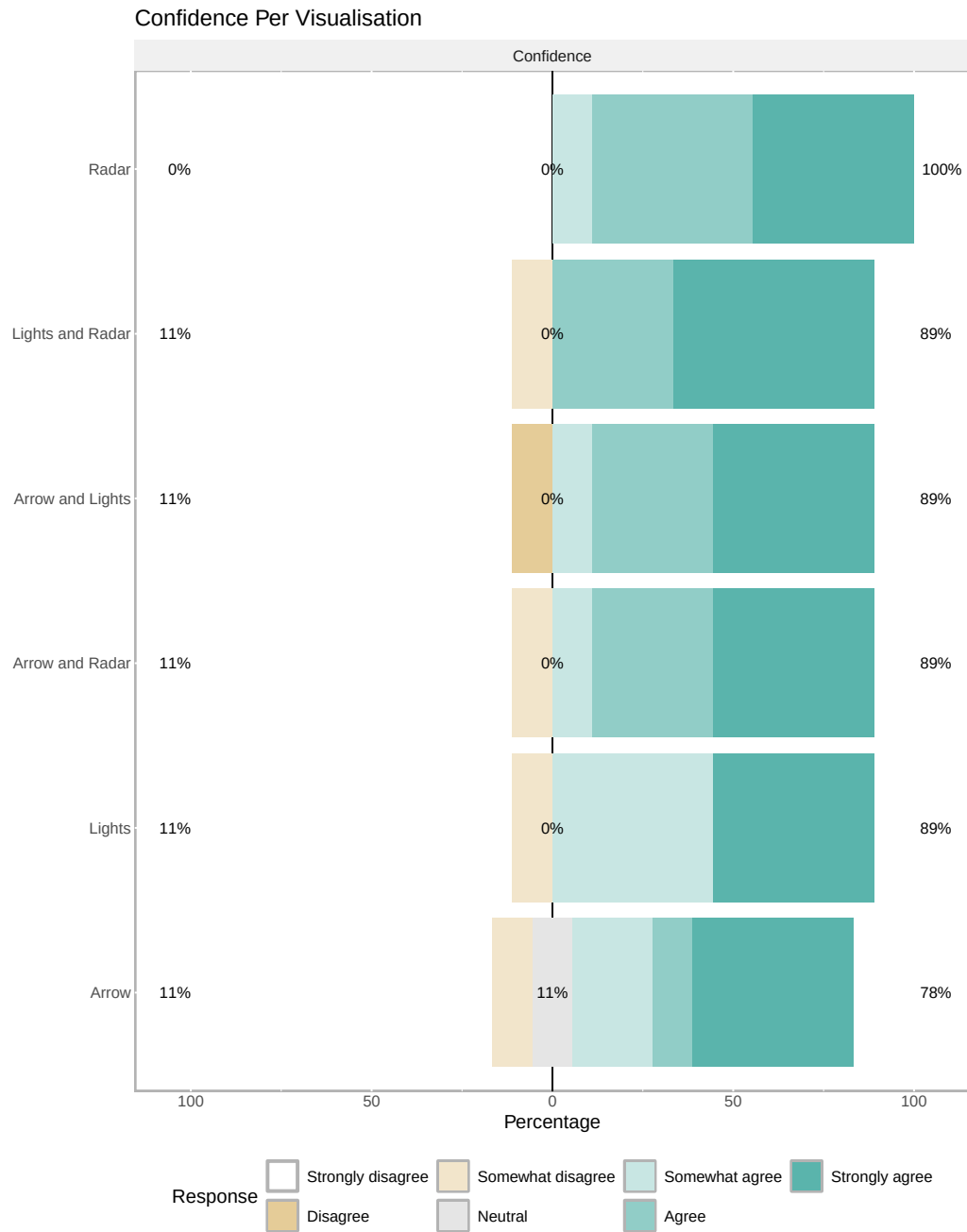


Figure B.2: Plot of Likert scale responses to the survey question "I felt confident using this visualisation to localise sound." Shows the majority of responses were strongly agree for all visualisations

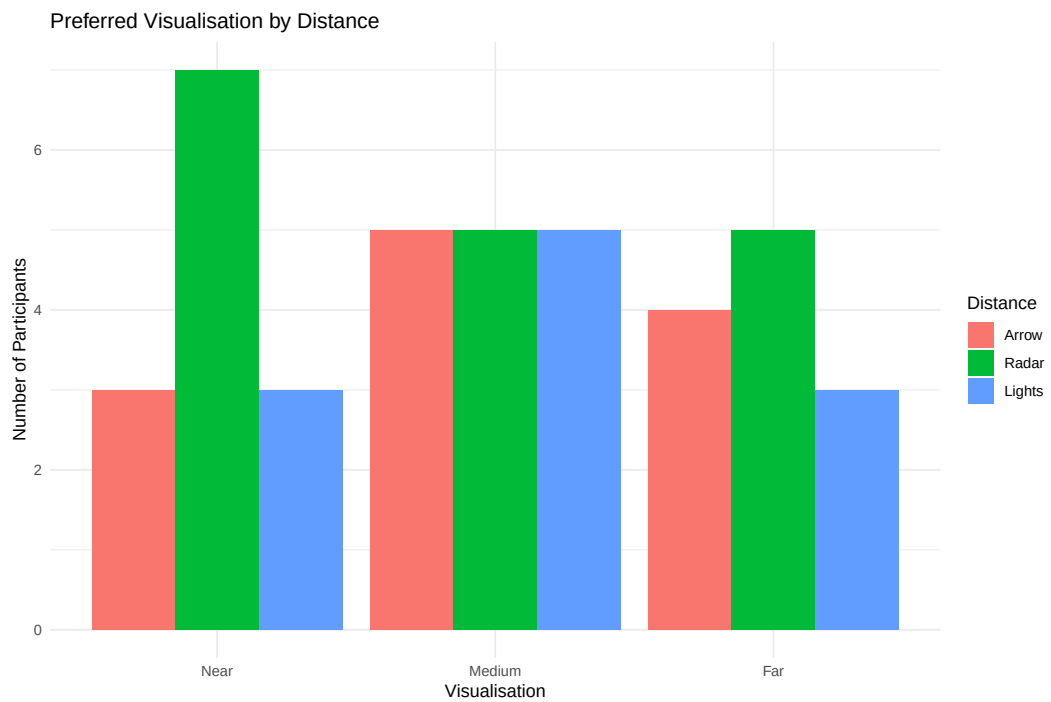


Figure B.3: Bar chart of preferred visualisations by participants at near, medium and far distance. The combinations are split into individual visualisations giving three visualisations, Arrow, Radar and Lights. Radar has the highest count for near, all visualisations have the same count for medium and Radar has the highest count for far followed by Arrow.

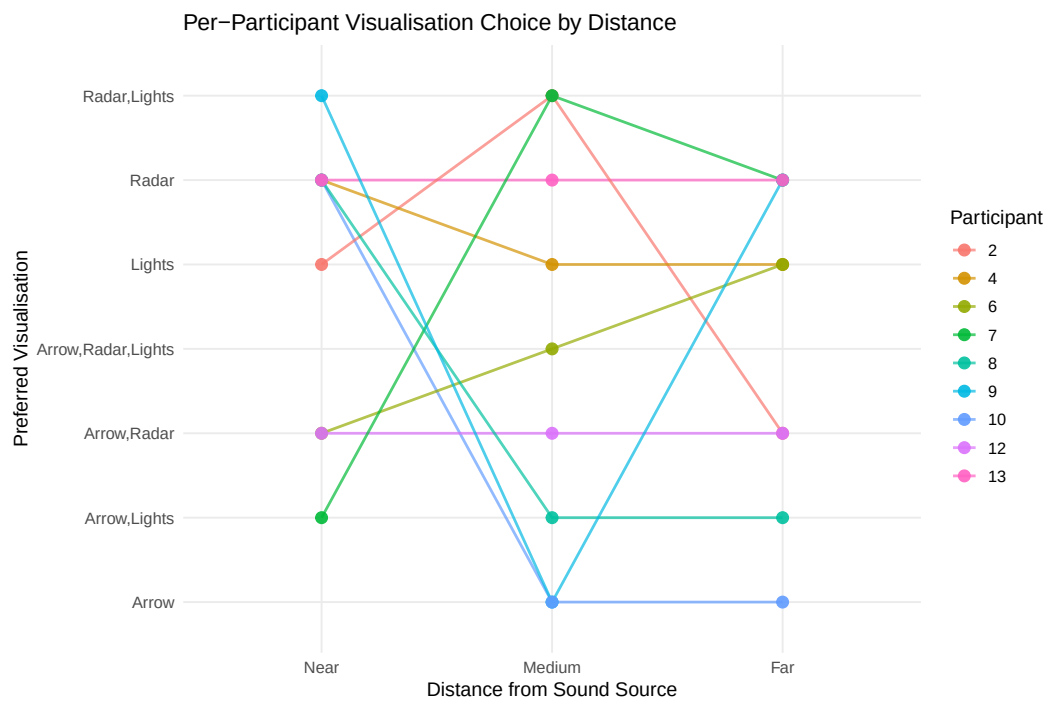


Figure B.4: Plot of participant visualisation choice at each distance. Horizontal lines indicate participant chose the same visualisations across all distances, uneven lines represent participants choosing different visualisations at different distances.

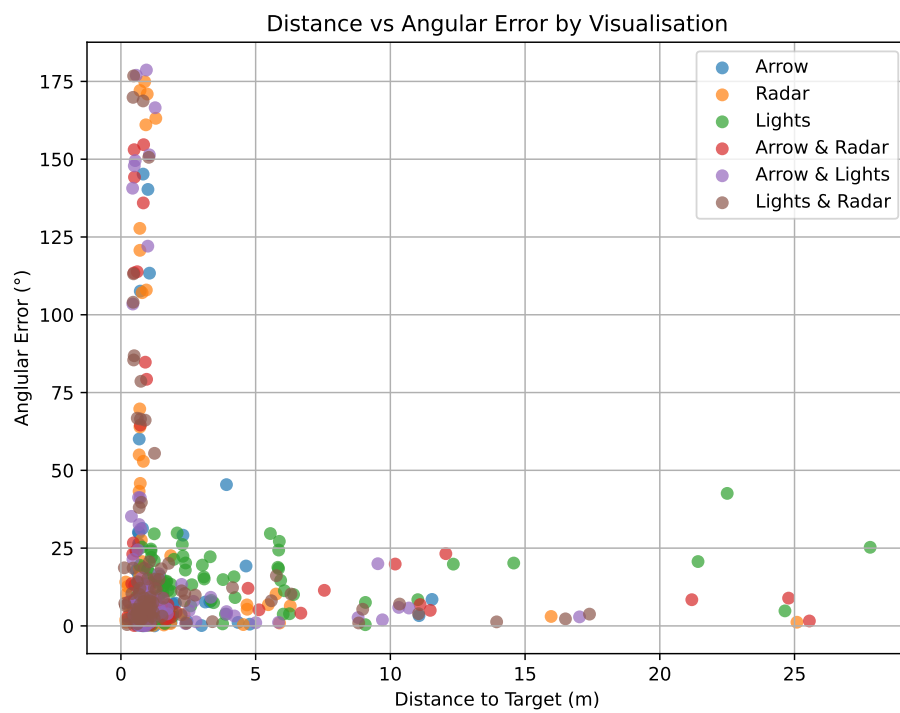


Figure B.5: Angular error plotted against Distance to Target, using colour to encode visualisation type. It is clear that as the distance decreases towards under 2.5 metres the angular error increases drastically ranging from 0 to 180.

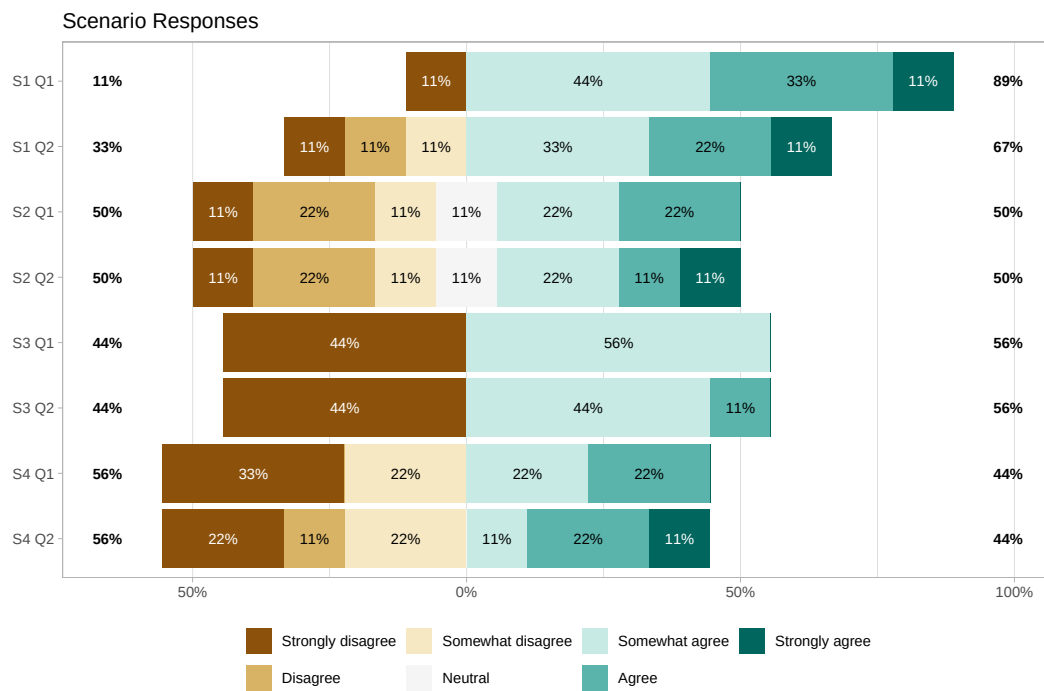


Figure B.6: Plot of Likert responses to questions asking participants how useful they thought the system would be in the scenario (Q1) and if they would use it (Q2). Scenario 1 has the highest percentage of respondents agreeing with 89% while the rest of the scenarios have around a 50/50 split between agreeing and disagreeing.

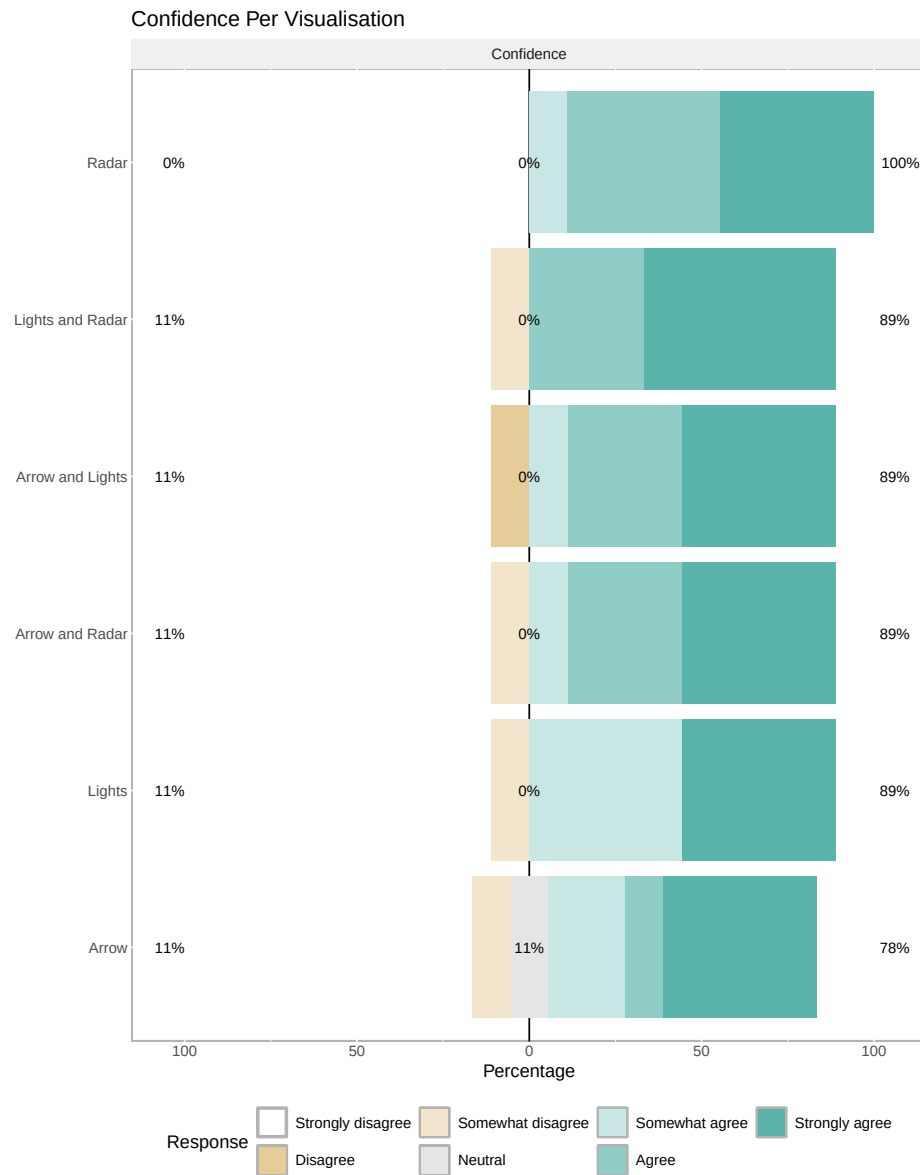


Figure B.7: Plot of Likert scale responses to the survey question "I felt confident using this visualisation to localise sound." Shows the majority of responses were strongly agree for all visualisations

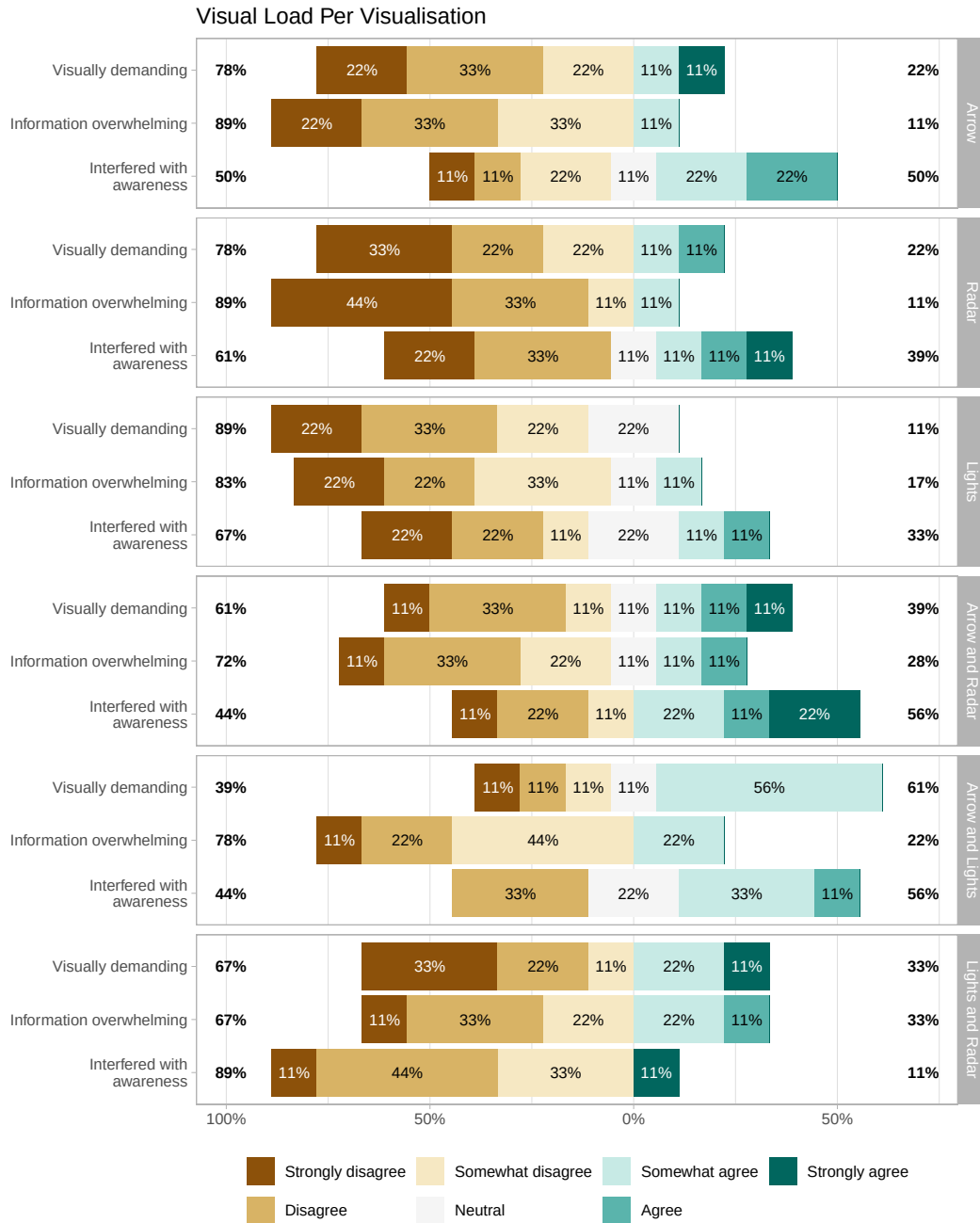


Figure B.8: Plot of Likert scale responses to questions on visual load for each visualisation.

C | Survey questionnaires and Interview Questions

Default Question Block

What is your Participant ID (PID)?

Please rank the visualizations from best to worst.

- Arrow
- Radar
- Lights
- Arrow and Radar
- Radar and Lights
- Lights and Arrow

Which visualizations would you prefer for different distances from the sound source?

	visualizations		
	Arrow	Radar	Lights
Near	☐	☐	☐
Medium	☐	☐	☐
Far	☐	☐	☐

For the following questions please consider the system being used with modern smartglasses capable of full Augmented Reality.

You will be asked to answer questions based on the following

scenarios:

Scenario 1: Working in a safety critical warehouse with hearing protection on.

Scenario 2: Walking along a busy road with headphones on.

Scenario 3: Responding to a fire alarm in a large building.

Scenario 4: Sitting in a cafe with headphones on.

	Strongly disagree	Disagree	Somewhat disagree	Neither agree nor disagree	Somewhat agree	Agree	Strongly agree
	1	2	3	4	5	6	7
The system would be useful in Scenario 1.				☒			
I would choose to use the system in Scenario 1.				☒			
The system would be useful in Scenario 2.				☒			
I would choose to use the system in Scenario 2.				☒			
The system would be useful in Scenario 3.				☒			
I would choose to use the system in Scenario 3.				☒			
The system would be useful in Scenario 4.				☒			
I would choose to use the system in Scenario 4.				☒			

Visualization Question Block

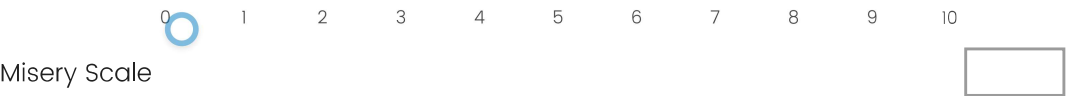
What is your Participant ID (PID)?

Which visualization have you just completed?

- ☐ Arrow
- ☐ Radar
- ☐ Lights
- ☐ Arrow and Radar
- ☐ Arrow and Lights
- ☐ Lights and Radar

Misery Scale (MISC)

Symptoms		MISC
No problems		0
Some discomfort, but no specific symptoms		1
Dizziness, cold/warm, headache, stomach/throat awareness, sweating, blurred vision, yawning, burping, tiredness, salivation, . . . but no nausea	Vague	2
	Little	3
	Rather	4
	Severe	5
	Little	6
Nausea	Rather	7
	Severe	8
	Retching	9
Vomiting		10



NASA TLX for this visualization.

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20

11

11



11

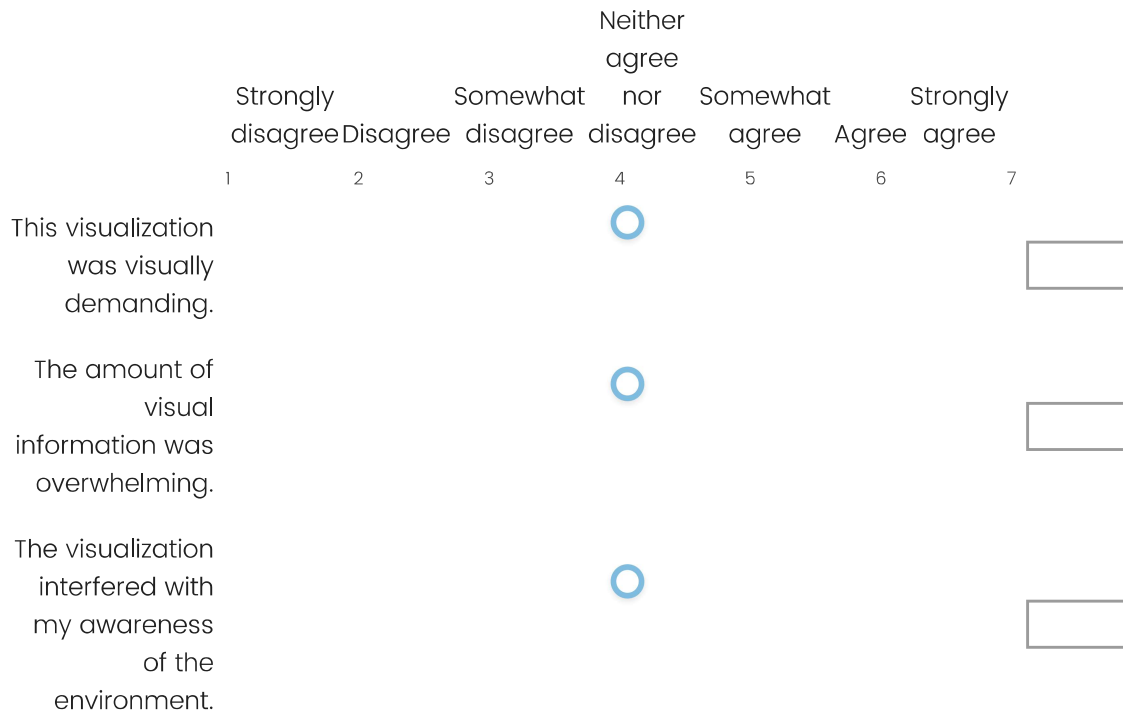
11

11

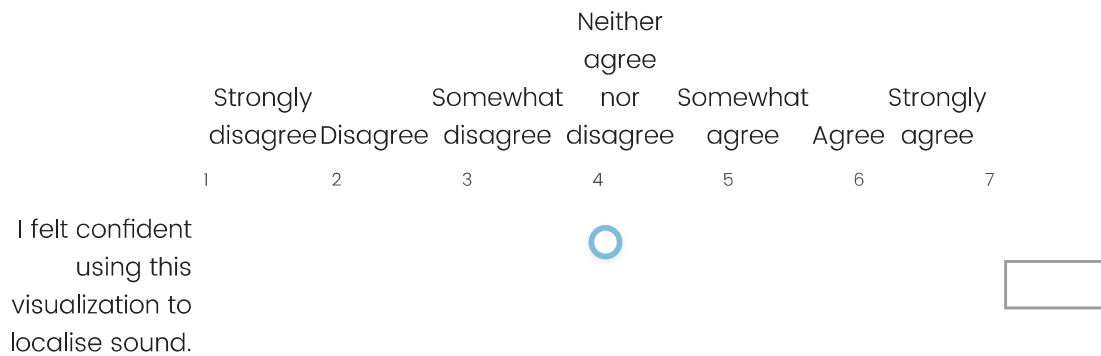


11

Visual load.



Confidence.



Did you find anything confusing about this visualization?

Any suggestions for improving this visualization?

Powered by Qualtrics

Interview Questions.

- 1.) Why did you rank X as the best visualization?
- 2.) Why did you rank X as the worst visualization?
- 3.) You chose visualization X for far, Y for medium and Z for far distance - what was it about each that made you choose them for that distance?
- 4.) Would you be likely to use this system in your everyday life if it were implemented into a pair of AR smart glasses?
 - a.) Why or why not?
 - b.) When or where in your everyday life would you use it?
- 5.) Would using this system make you feel more confident wearing headphones or other devices that may impair your hearing.
- 6.) Given a directional microphone, what else would you want to use it for to enhance and or augment your perception of your aural surroundings?
- 7.) Are there any scenarios where you think this system may be distracting and not appropriate?
- 8.) Do you have any suggestions for improving the system overall or for an actual visualization?
- 9.) Who do you think would benefit most from this system?

D | Manual

Manual

/code

This folder contains python scripts that can be run to estimate the location of spatial sounds using the miniDSP UMA 8 Microphone array. These scripts also broadcast these measures on the connected network using a sockets server.

`OnBoardDoa.py` use's the UMA-8's onboard DOA method.

`SPRPHATDOA.py` is the DOA method implemented in this project which uses the SPR-PHAT DOA method using the UMA-8's RAW mode.

/UnityProject

This folder contains the unity project used for the wearable prototype and experiment.

All scripts, scenes and assets used, such as images and materials, are stored in the subdirectory `/Assets`

The prototype scene is located at `/Assets/Scenes/Passthrough_demo.unity`

The experiment scene is located at `Assets/Scenes/VR.unity`

The scripts used for each visualisation technique can be found in `Assets/Scripts`

Build instructions

To run the prototype

- Build the unity project to a quest 3 headset with all scenes included.
- Connect the UMA-8 to a laptop, making sure the UMA-8 is running in the RAW advanced mode.
- Run `pip install requirements.txt` to install dependencies.
- Run `python SPRPHATDoa.py` in the code folder to begin networking sound source locations to the headset.
- Run the project on the quest 3 and hold down both primary buttons to boot into the prototype scene.

To run the experiment

- Build the unity project to a quest 3 headset.
- Open the project on the headset.
- press the primary button on the right controller to log metrics and move onto next trial.
- press the primary button on the left controller to swap visualisation.
- press the menu button on the left controller to start the experiment.
- After 10 trials a break will begin, press the left menu button to start the next condition.
- Once the experiment has started all log data will be saved to the headset.

Requirements

- Python 3.9
- Unity 2022.3.XX LTS
- miniDSP UMA-8 USB mic array - V2.0
- Packages: listed in `requirements.txt`
- Tested on Windows 10 and Windows 11

Test steps

Run `SPRPATDoa.py` in the code folder with the UMA-8 in RAW advanced mode to check that the script is working.

Build and Run the Unity project to a quest 3 headset to check the project can build.

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